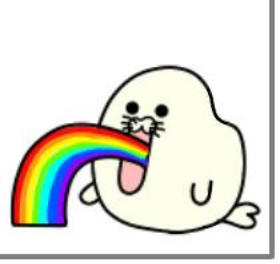
AZAYAKA

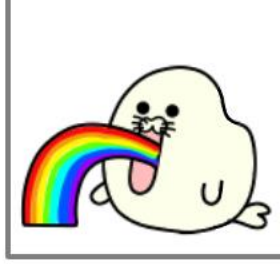
数式とジオメトリ
Version 0.1

Azayaka for Document

安井　秀輔

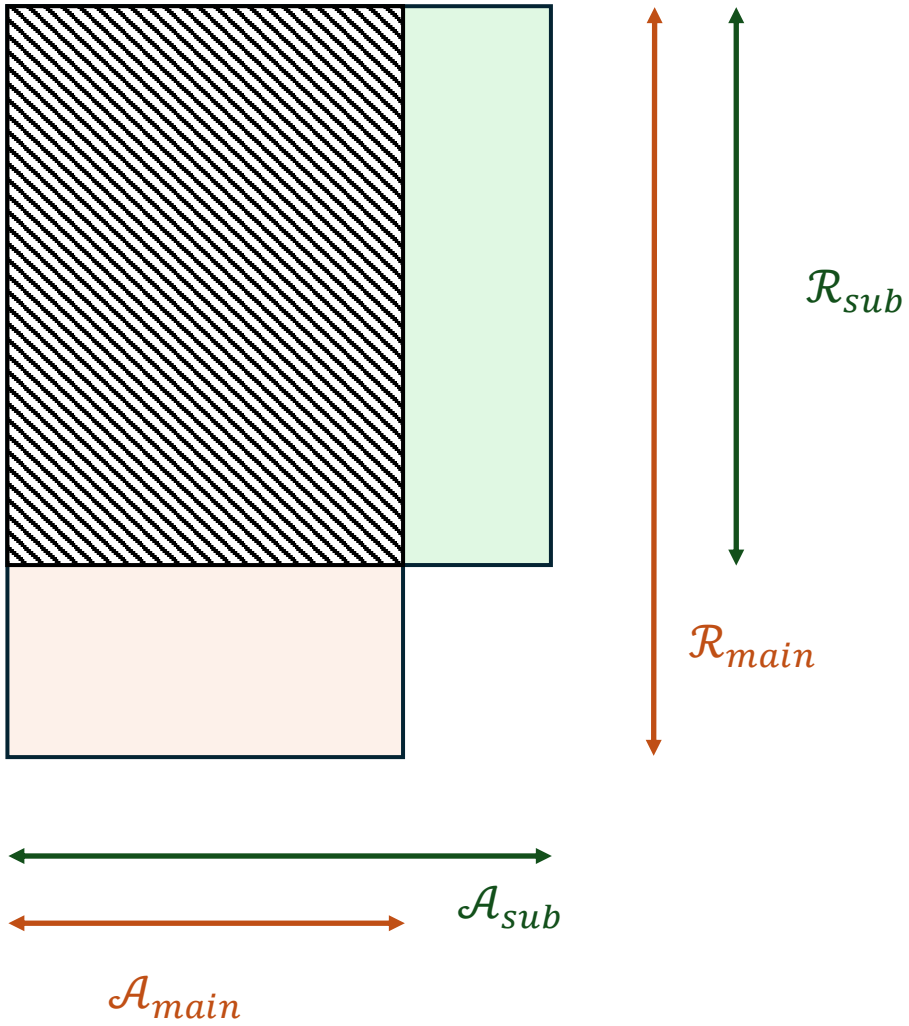


Equation Condition No 1.1



メイン軌道

サブ軌道



記号	意味
$\mathcal{R}_{main}$	メイン軌道の レンジサンプル数
$\mathcal{R}_{sub}$	サブ軌道の レンジサンプル数
$\mathcal{A}_{main}$	メイン軌道の アジマスサンプル数
$\mathcal{A}_{sub}$	サブ軌道の アジマスサンプル数

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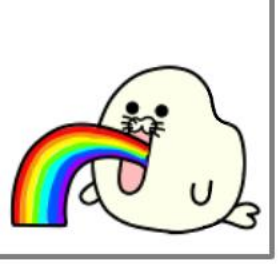
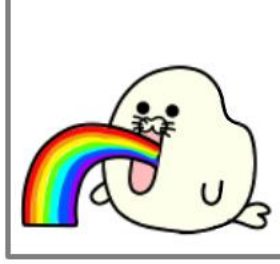
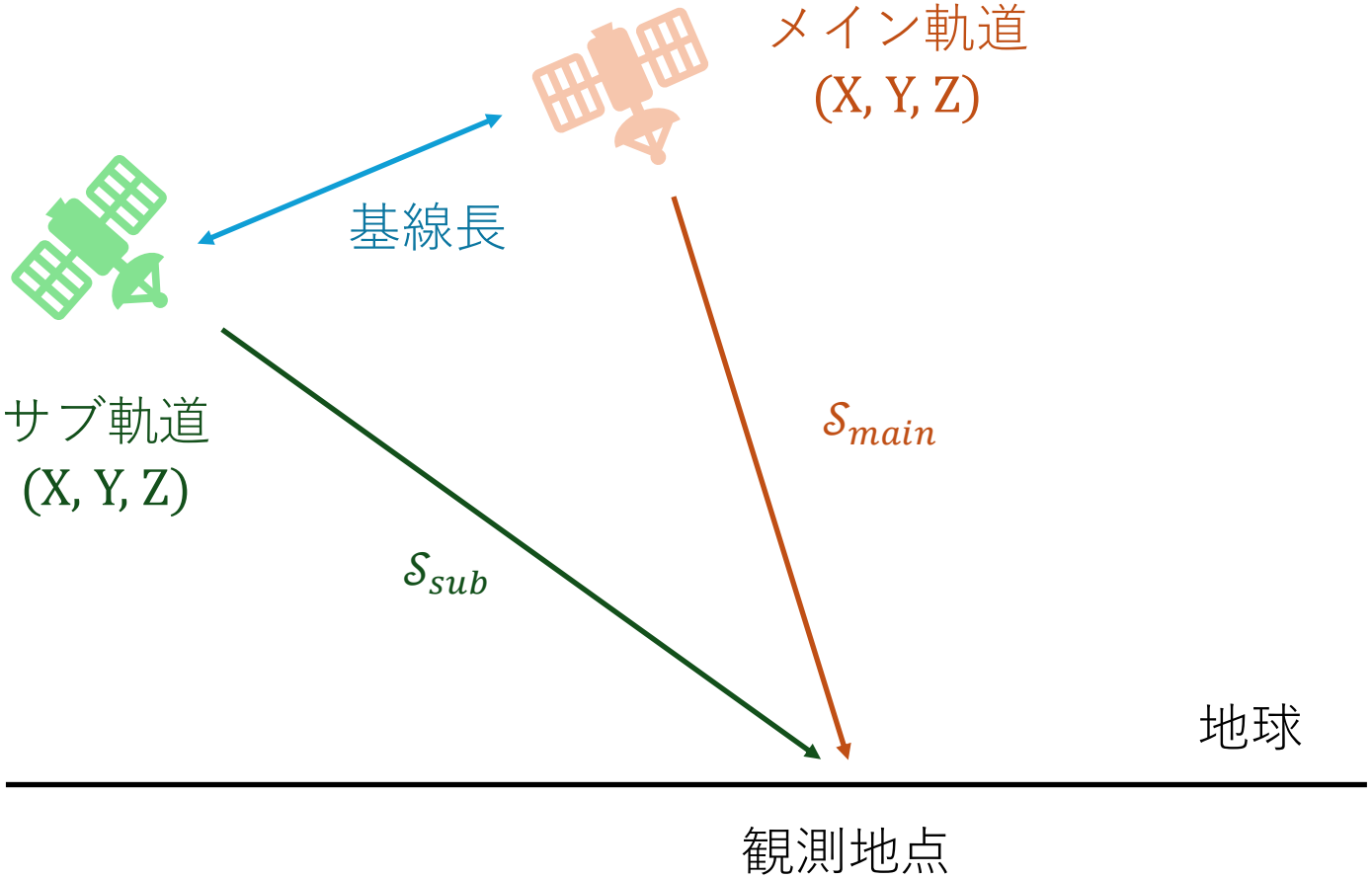


Equation Condition No 1.2





記号	意味
S_{main}	メイン軌道のスラントレンジ
S_{sub}	サブ軌道のスラントレンジ
(X, Y, Z)	ECEF座標の3次元軸

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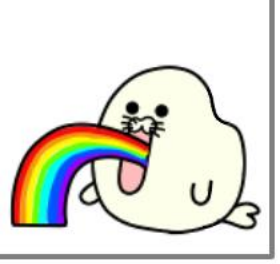
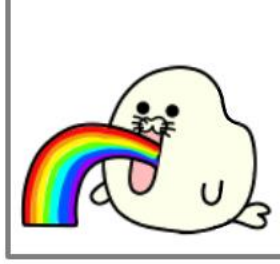
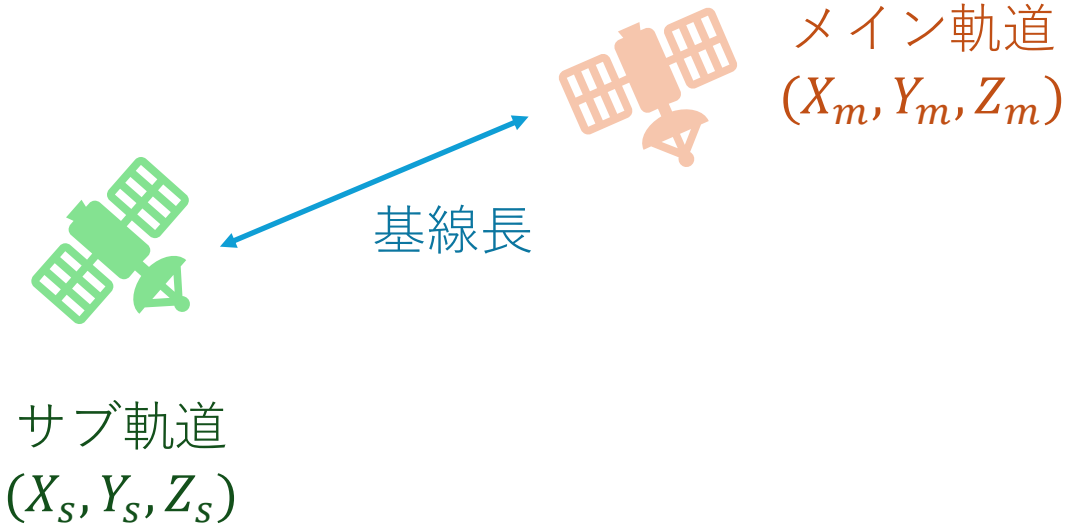


Equation Condition No 1.2





記号	意味
(X_m, Y_m, Z_m)	メイン軌道の ECEF座標の3次元軸
(X_s, Y_s, Z_s)	メイン軌道の ECEF座標の3次元軸

$$\text{基線長}\ B := \sqrt{(X_m - X_s)^2 + (Y_m + Y_s)^2 + (Z_m - Z_s)}$$

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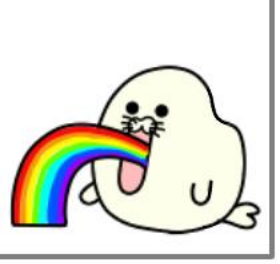
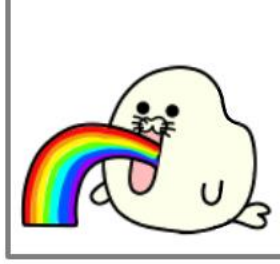
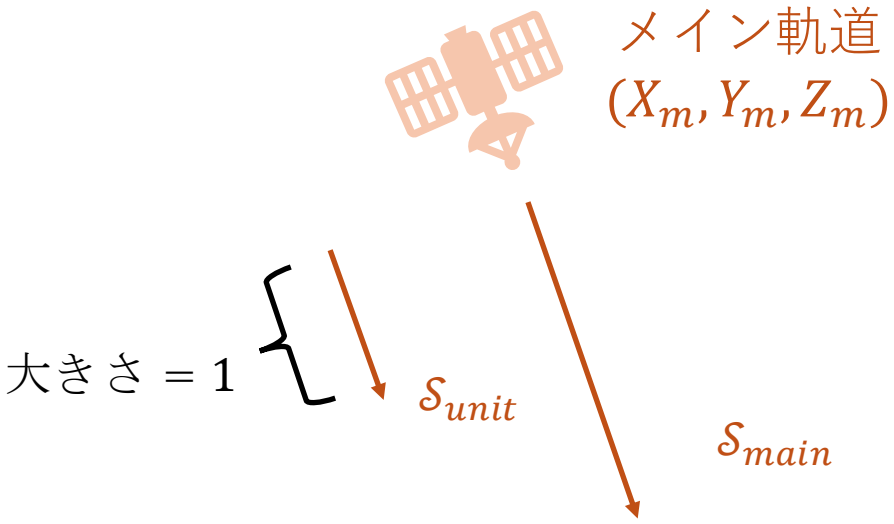


Equation Condition No 1.3





記号	意味
S_{unit}	S_{main}の単位ベクトル

$$\text{単位ベクトル}S_{unit} = \frac{S_{main}}{|S_{main}|} = \frac{(X_m, Y_m, Z_m)}{(X_m^2, Y_m^2, Z_m^2)}$$

$$|S_{unit}| = 1$$

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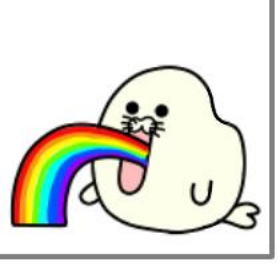
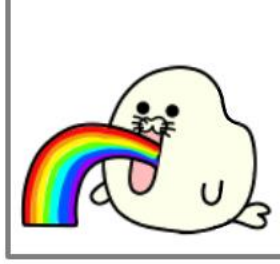
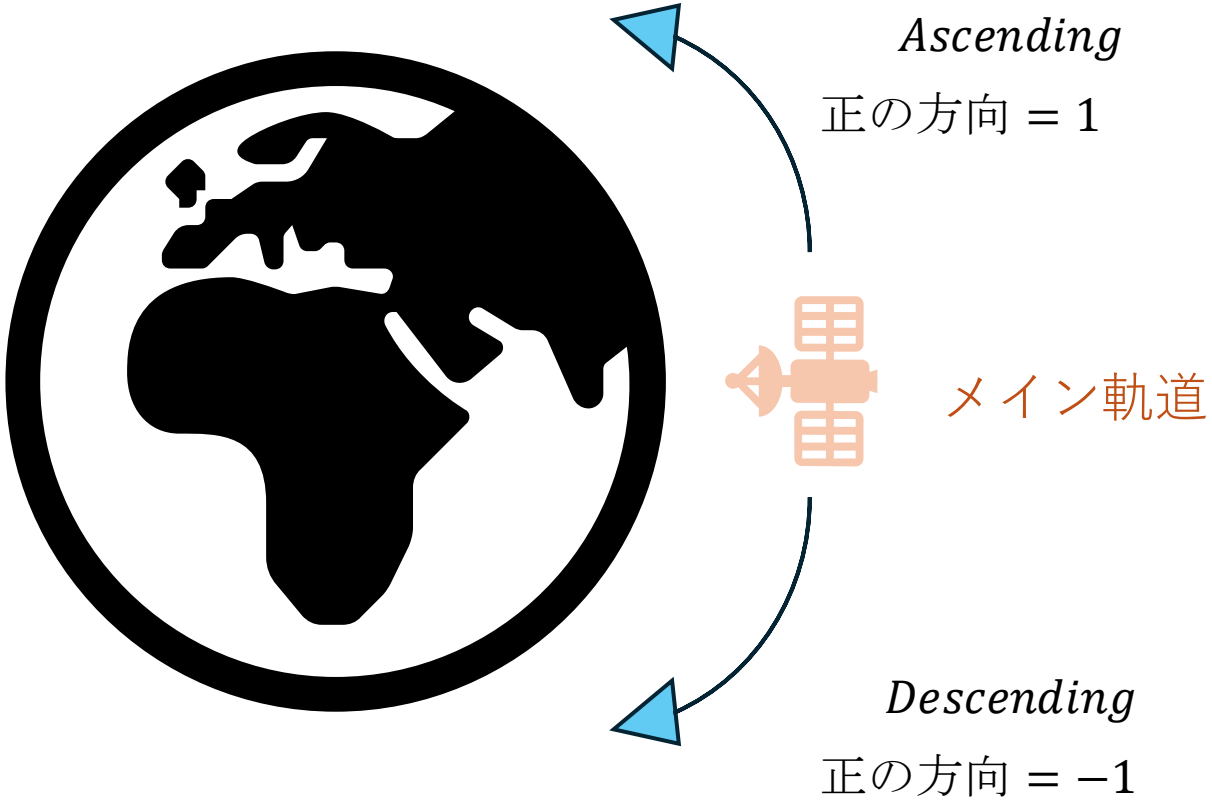


Equation Condition No 1.4





記号	意味
Ascending	**北方方向の軌道**
Descending	**南方方向の軌道**
符号	**メイン軌道が 北方か南方か**

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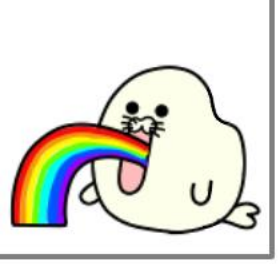

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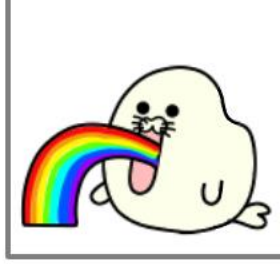

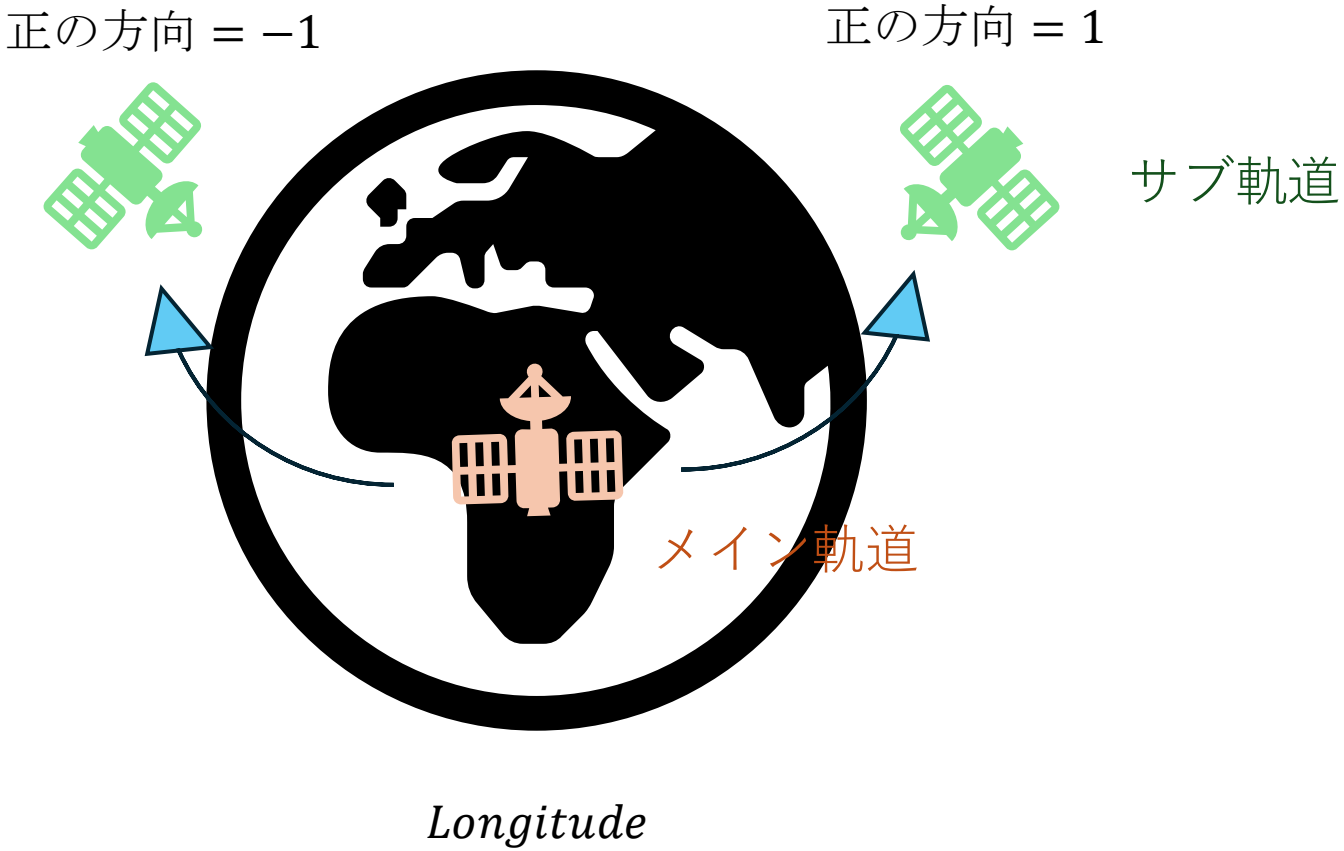

記号	意味
Longitude	地球を東西に分割した角度
符号	メイン軌道に対して、 サブ軌道が西か東か

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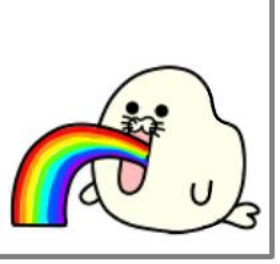

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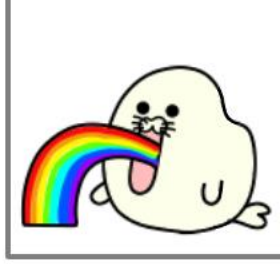

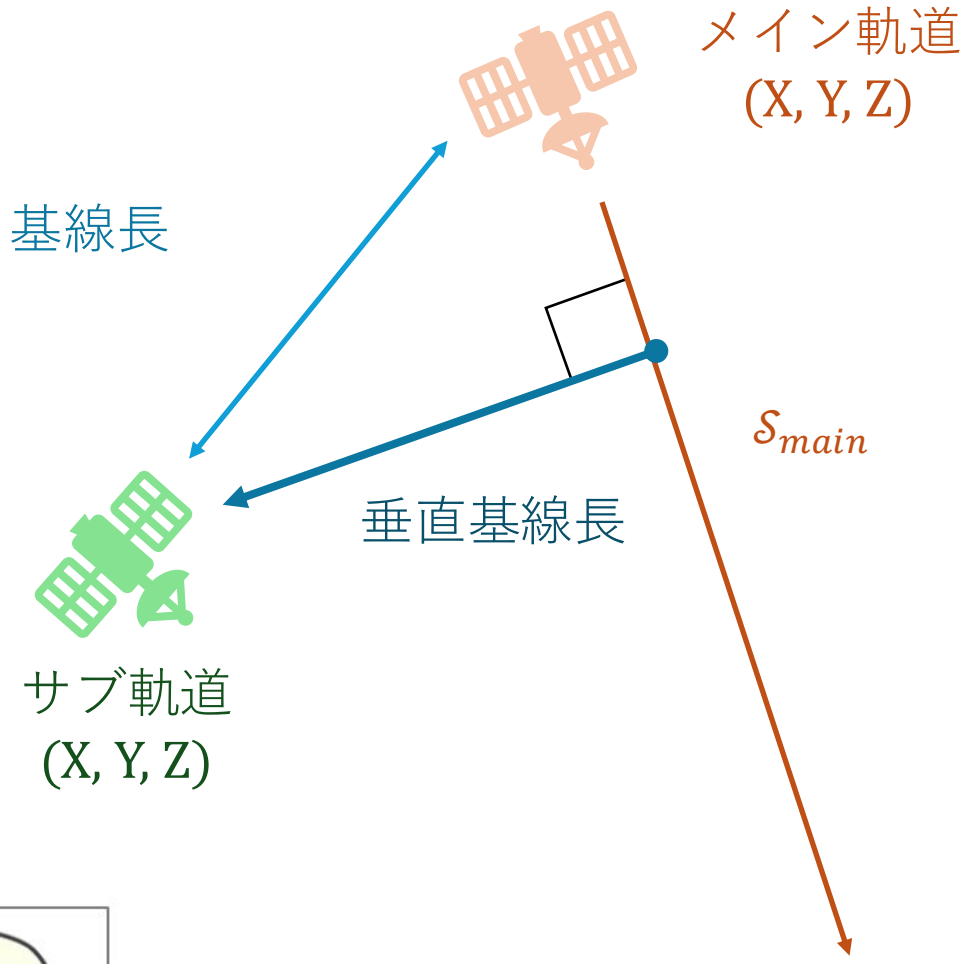

記号	意味
基線長 (*Baseline*)	メイン軌道とサブ軌道の差 干渉解析の基準の１つ
垂直基線長 (*Baseline Perpendiculer*)	ベースラインの垂直成分

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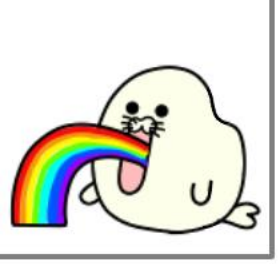

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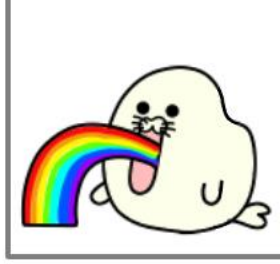

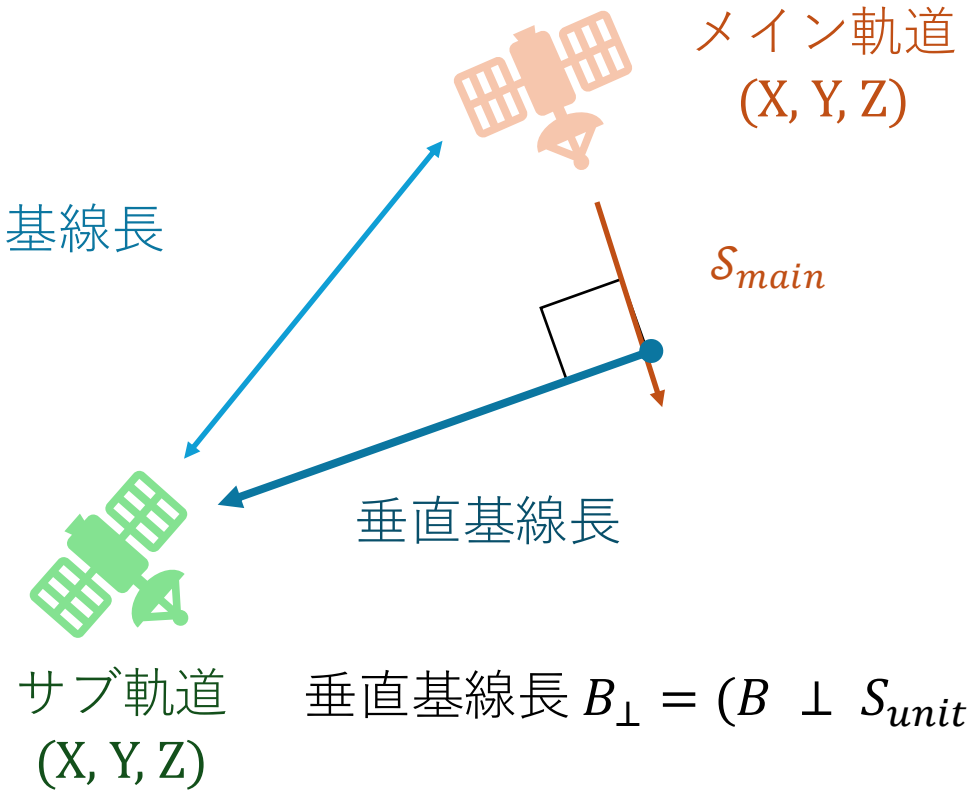

垂直基線長 $B_{\perp} = (B \perp S_{unit})$

記号	意味
$B_{\perp}$	垂直ベースライン
S_{unit}	S_{main}の単位ベクトル

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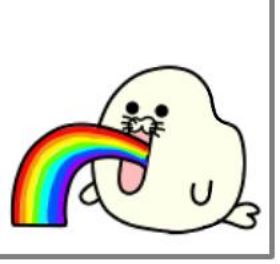
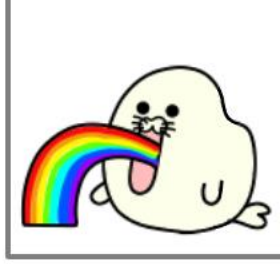
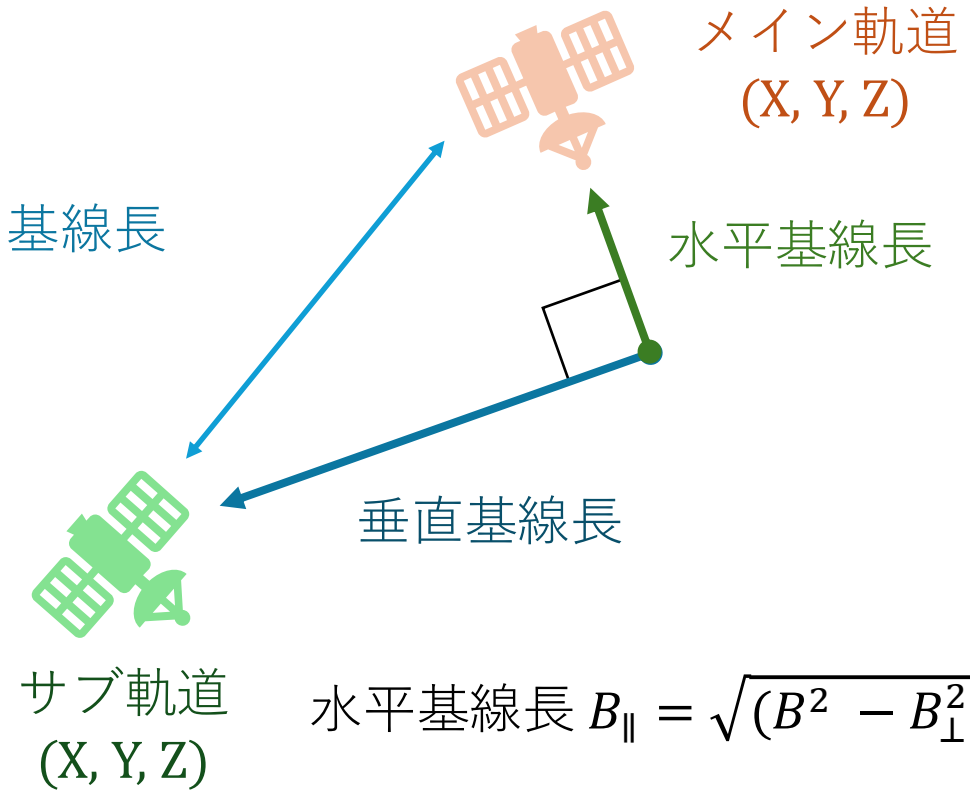


Equation Condition No 1.6





水平基線長 $B_{\parallel} = \sqrt{(B^2 - B_{\perp}^2)}$

記号	意味
B	ベースライン
$B_{\perp}$	ベースラインの垂直成分
$B_{\parallel}$	ベースラインの水平成分

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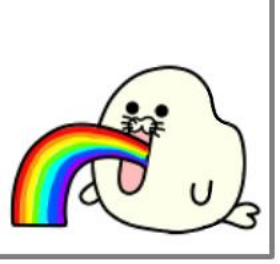
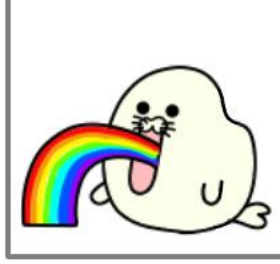
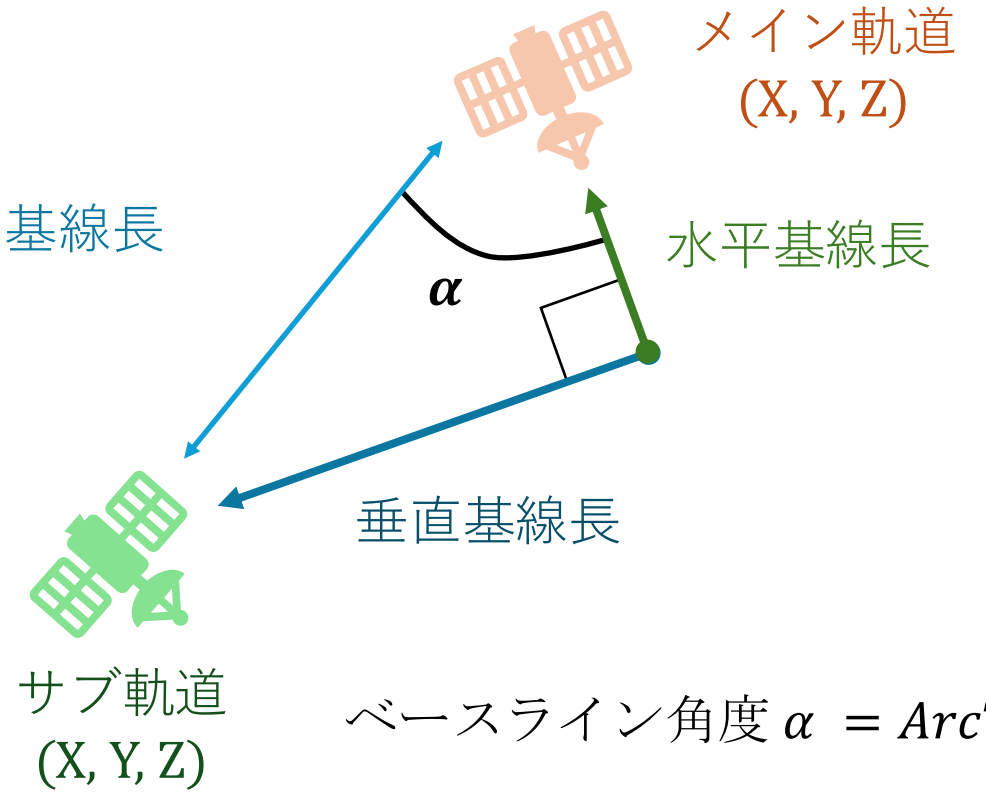


Equation Condition No 1.7





ベースライン角度 $\alpha = ArcTan(\frac{B_{\perp}}{B_{\parallel}})$

記号	意味
α	垂直ベースラインと水平ベースラインの逆正接の角度
B	ベースライン
$B_{\perp}$	ベースラインの垂直成分
$B_{\parallel}$	ベースラインの水平成分

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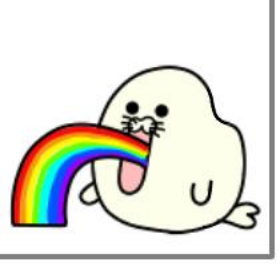
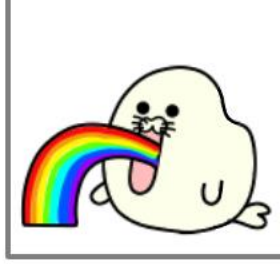
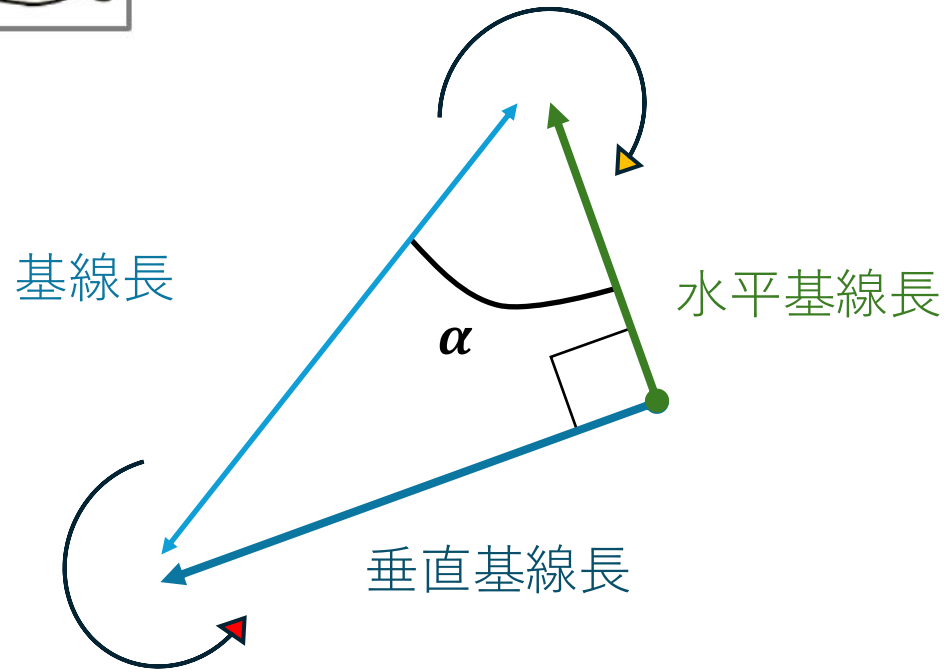


Equation Condition No 1.8





三角関数: $\sin(\alpha)$, $cos(\alpha)$

記号	意味
α	垂直ベースラインと水平ベースラインの逆正接の角度
B	ベースライン
$B_{\perp}$	ベースラインの垂直成分
$B_{\parallel}$	ベースラインの水平成分

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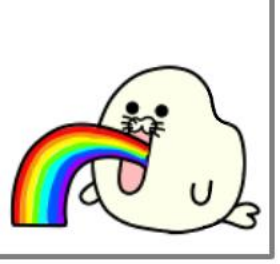
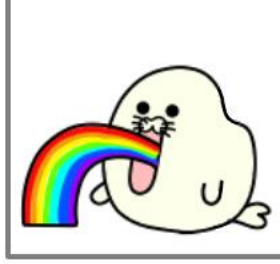
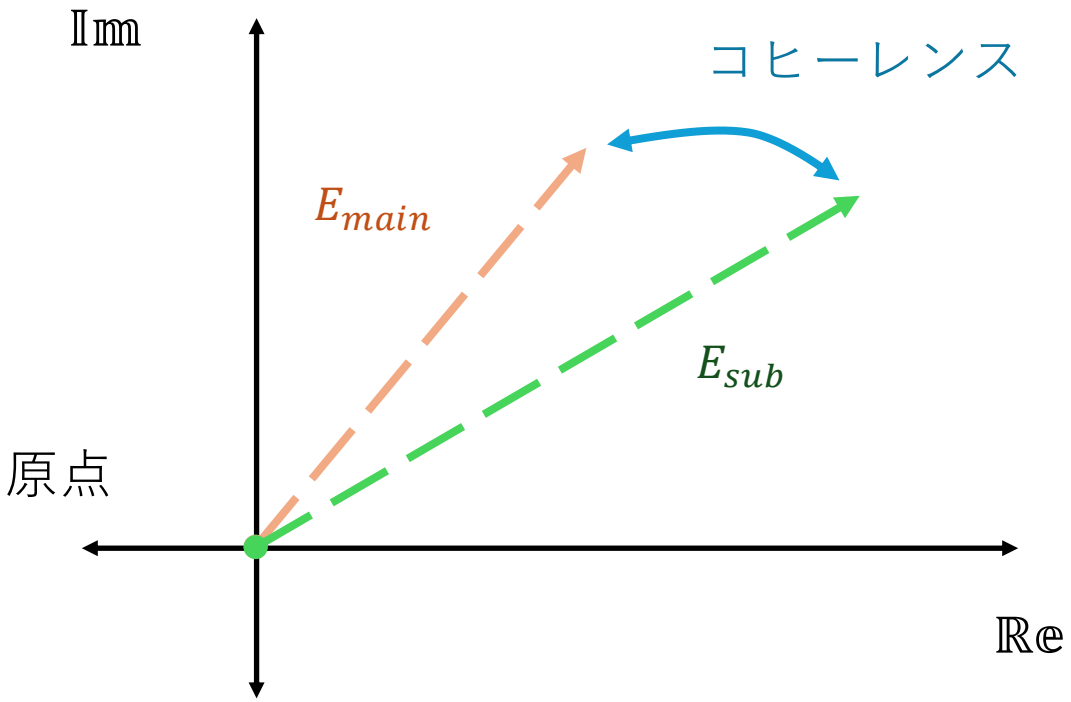


Equation Condition No 1.10



複素平面



$$Coherence := \frac{\left|\sum_{n=0}^{k}(E_{main}\cdot\overline{E_{sub}})\right|}{\sqrt{\left(\sum_{n=0}^{k}|E_{main}| * \sum_{n=0}^{k}|E_{sub}|\right)}}$$

記号	意味
$\lvert X\rvert$	X の大きさ
E_{main}	メイン軌道の反射信号
E_{sub}	サブ軌道の反射信号
$\bar{z}$	z(複素数)の複素共役

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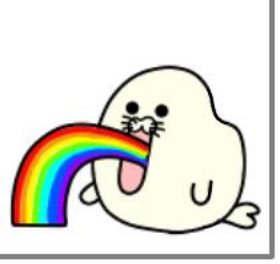
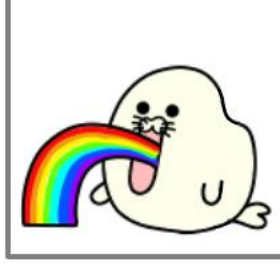
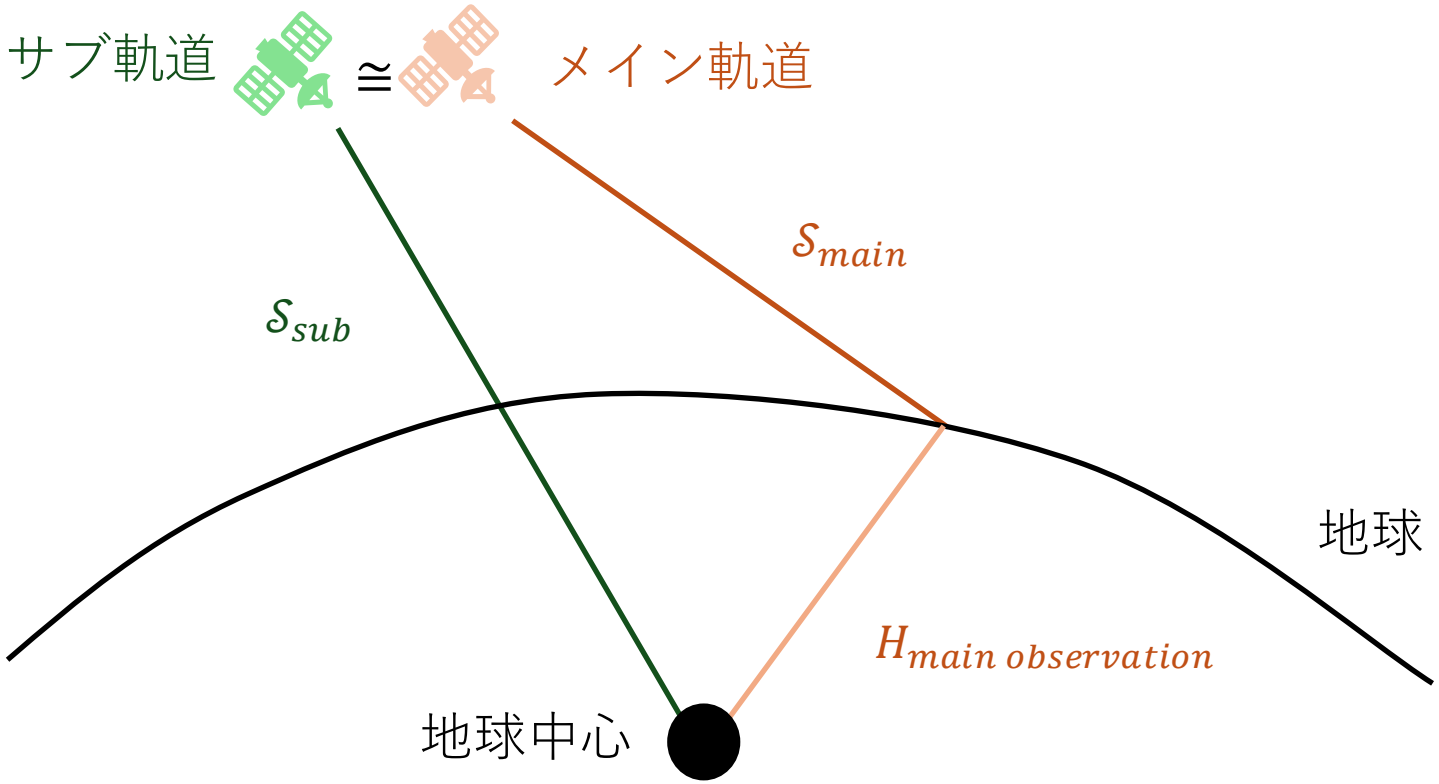


Equation Condition No 1.11





記号	意味
H_{sub}	サブ軌道の衛星高度
$H_{main\ observation}$	メイン軌道の観測地点までの距離
θ_{hight}	観測地点の地形を考慮した観測角度の違い

$$cos(\theta_{Height}) = \frac{S_{main}^2 + H_{sub}^2 - H_{main\ observation}}{2\ S_{main}\ H_{sub}}$$

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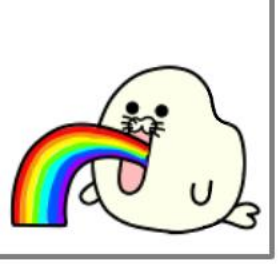
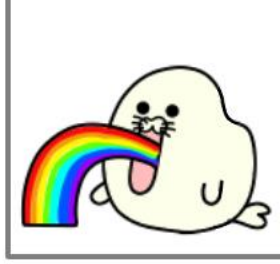
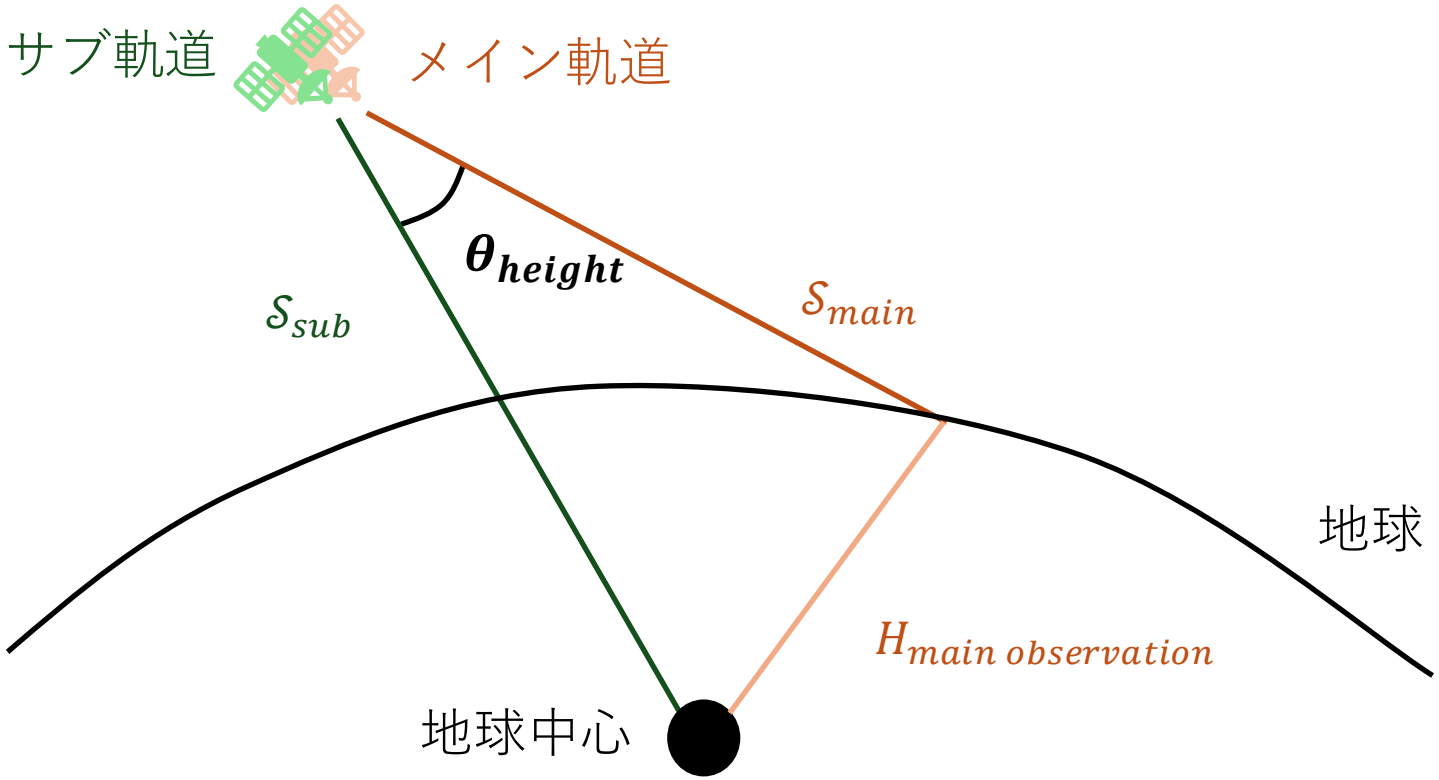


Equation Condition No 1.11





記号	意味
H_{sub}	サブ軌道の衛星高度
$H_{main\ observation}$	メイン軌道の観測地点までの距離
θ_{hight}	観測地点の地形を考慮した観測角度の違い

$$sin(\theta_{Height}) = \sqrt{1 - (\cos(\theta_{height}))^2}$$

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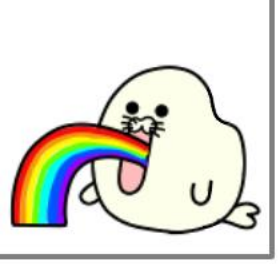
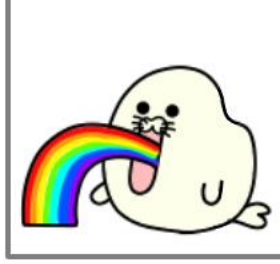
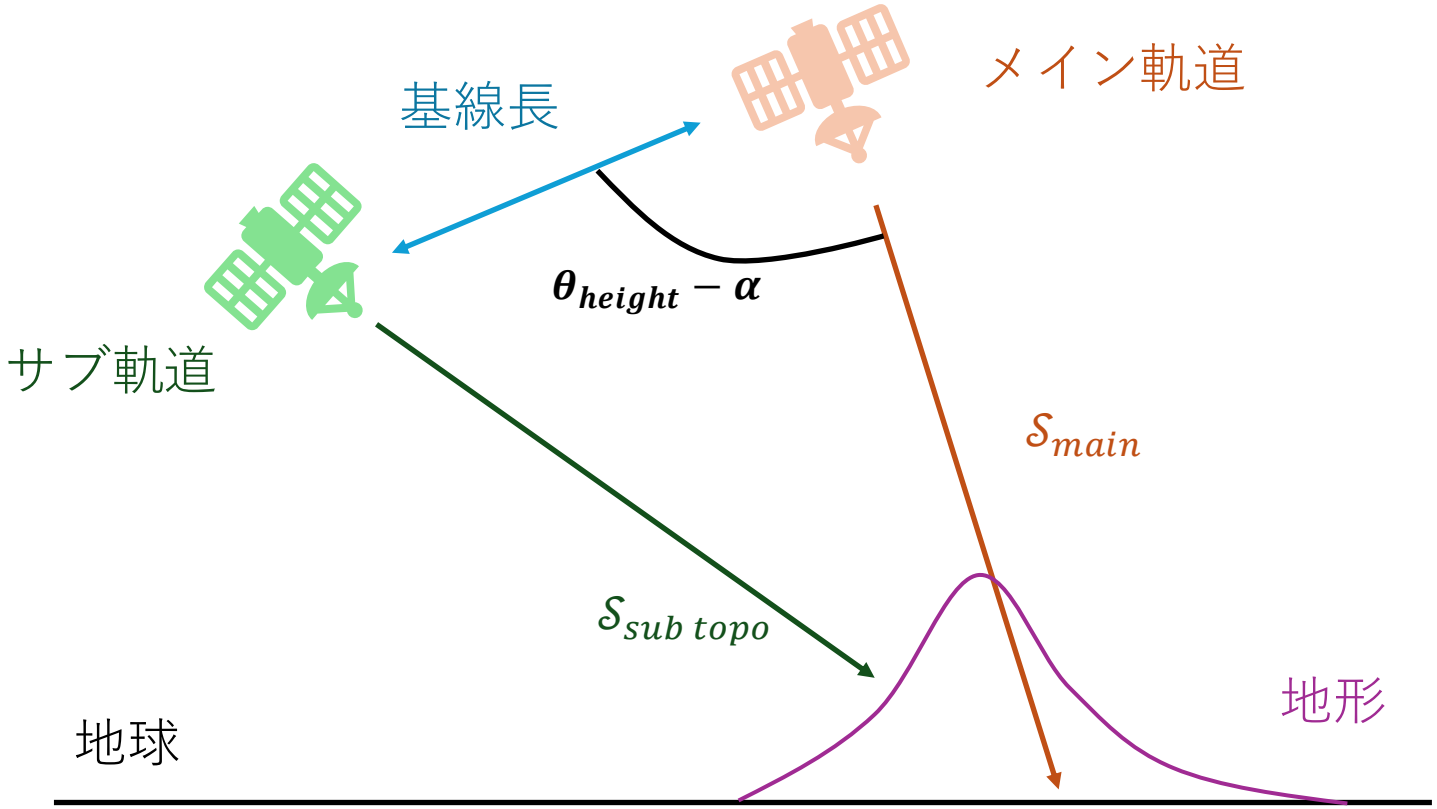


Equation Condition No 1.12



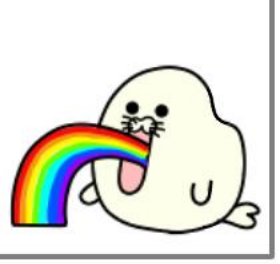


記号	意味
$S_{sub\ topo}$	サブ軌道からの地形を考慮したスラントレンジ
B	ベースライン
α	ベースライン角度
θ_{hight}	観測地点の地形を考慮した観測角度の違い

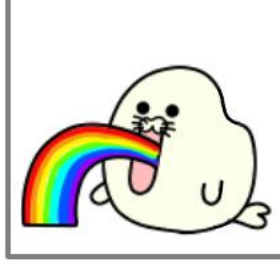
$$\text{地形二乗誤差}\ S_{sub\ topo}^2 = S_{main}^2 + B^2 - 2S_{main}B\sin(\theta_{height} - \alpha)$$

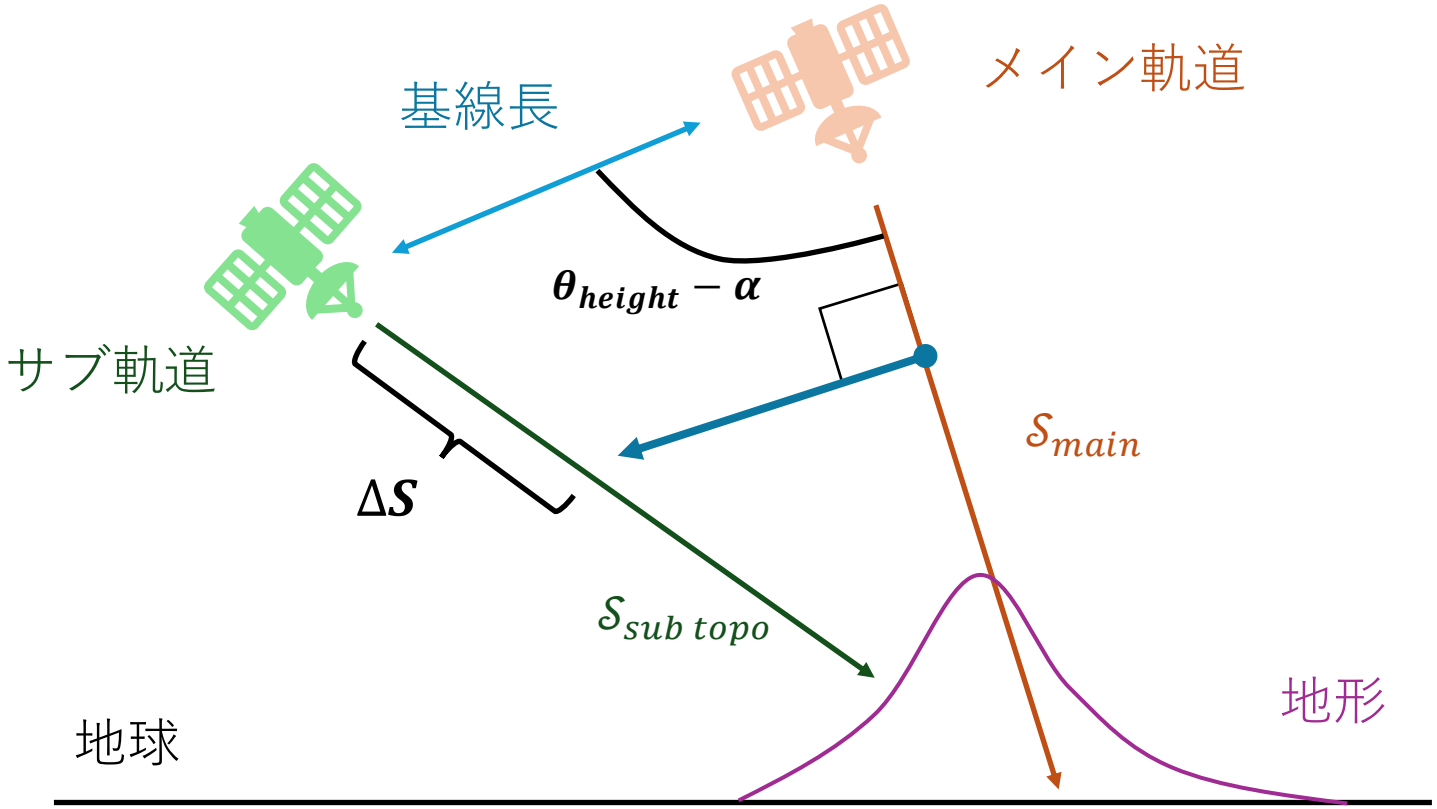
$$= S_{main}^2 + B^2 - 2S_{main}B\left(\sin(\theta_{height})\cos(\alpha) - \cos(\theta_{height})\sin(\alpha)\right)$$

© Azayaka



Equation Condition No 1.13





記号	意味
$S_{sub\ topo}$	サブ軌道からの地形を考慮したスラントレンジ
B	ベースライン
α	ベースライン角度
θ_{hight}	観測地点の地形を考慮した観測角度の違い
ΔS	軌道差によるスラントレンジの誤差

$$\text{位相誤差}\ \Delta S = \sqrt{S_{sub\ topo}^2} - S_{main}$$

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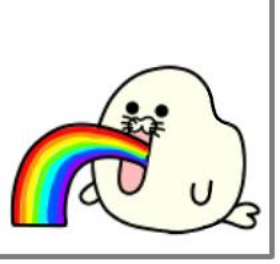
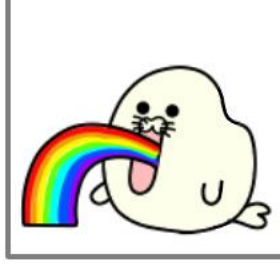
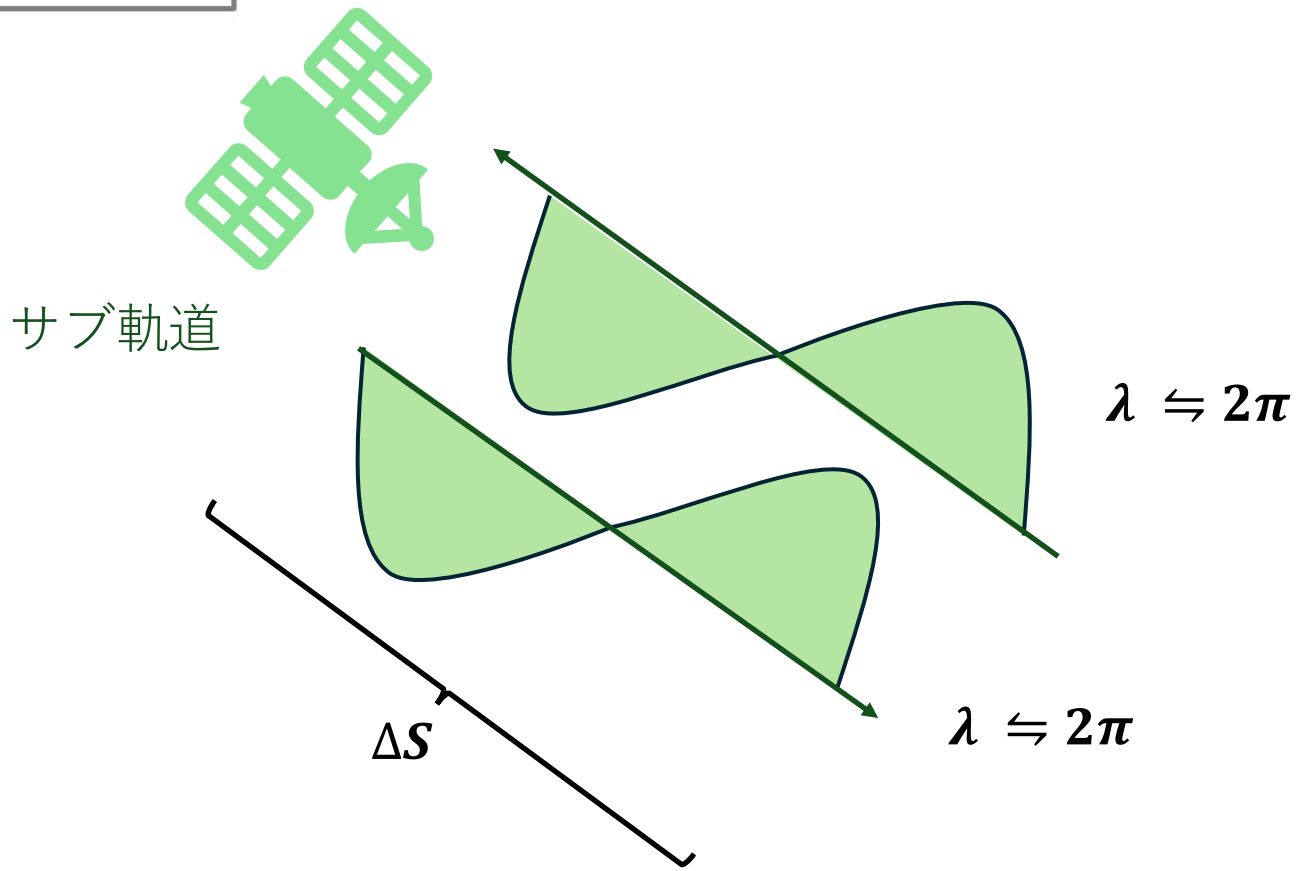


Equation Condition No 1.14





記号	意味
λ	波長
ΔS	軌道差によるスラントレンジの誤差
$exp(X)$	Xを変数とした指数関数
$\Delta\phi$	位相誤差

位相誤差 $\Delta\phi = \exp({}^{4\pi j}/_{\lambda\Delta S})$

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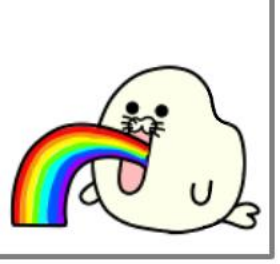
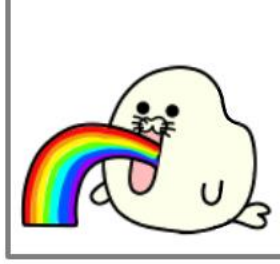
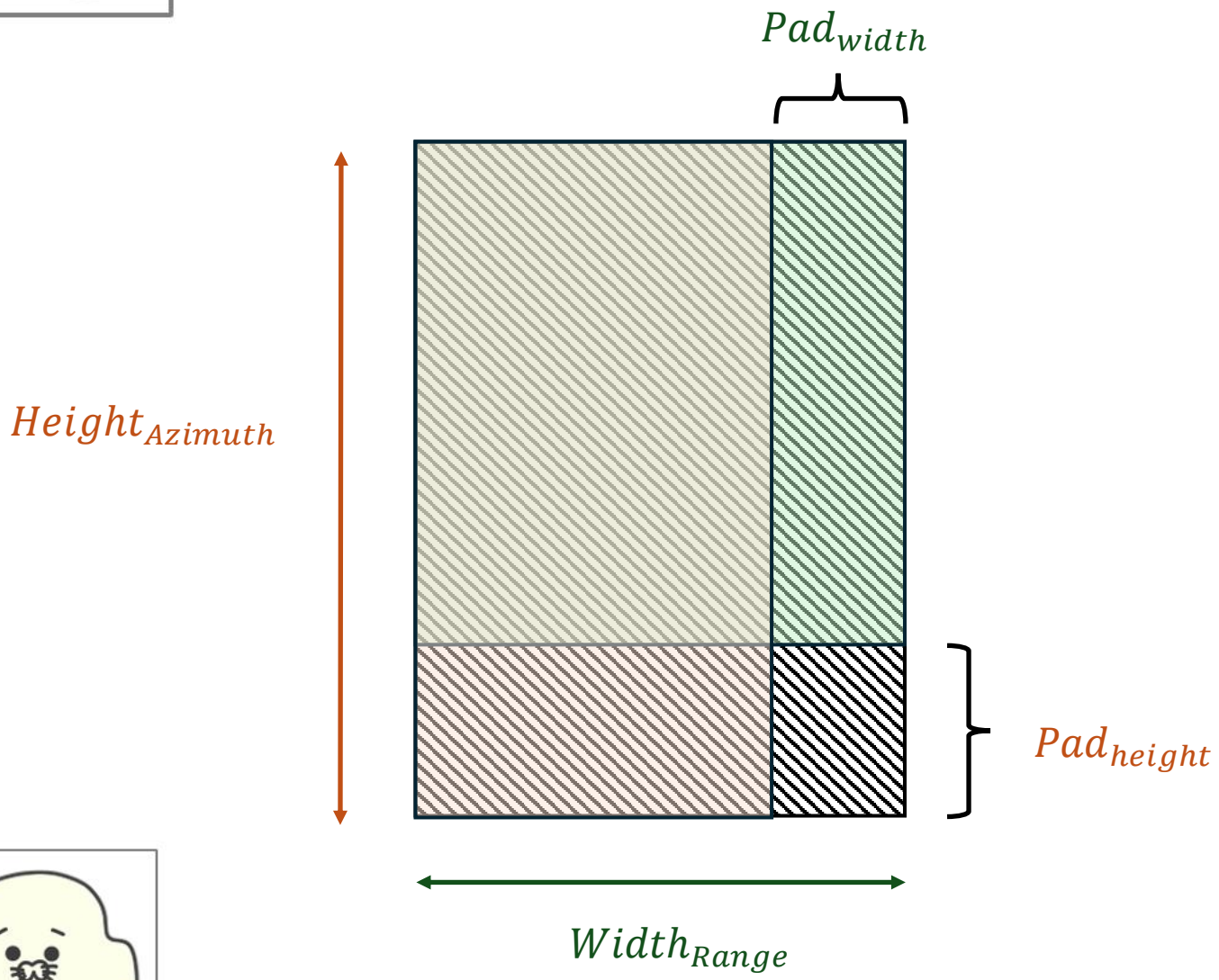


Equation Condition No 1.15





記号	意味
$Height_{Azimuth}$	画像の高さ(アジマス方向)
$Width_{Range}$	画像の横(レンジ方向)
Pad_{height}	拡張するピクセル数(高さ)
Pad_{width}	拡張するピクセル(横)

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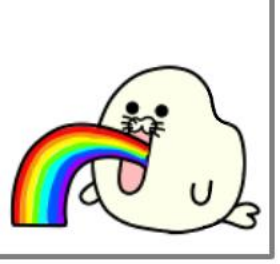
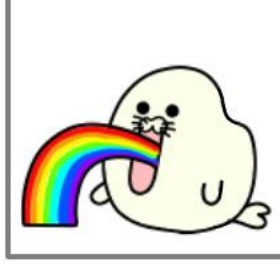


Equation Condition No 1.17



振幅とパワー比率

$$dB := 10\log_{10}(\frac{P}{P_0})$$

$$P \propto Amplitude^2$$

$$Intensity = 10\log_{10}(Amplitude^2) + P_{offset}$$
$$= 20\log_{10}(Amplitude) + P_{offset}$$

記号	意味
dB	デシベル値
$log_{10}(y)$	10を基底に持つ対数関数
P_i	パワー

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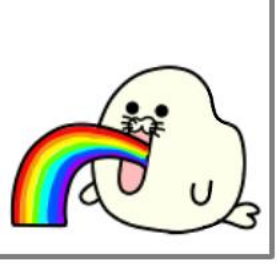
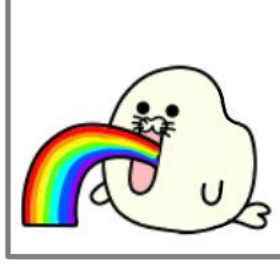
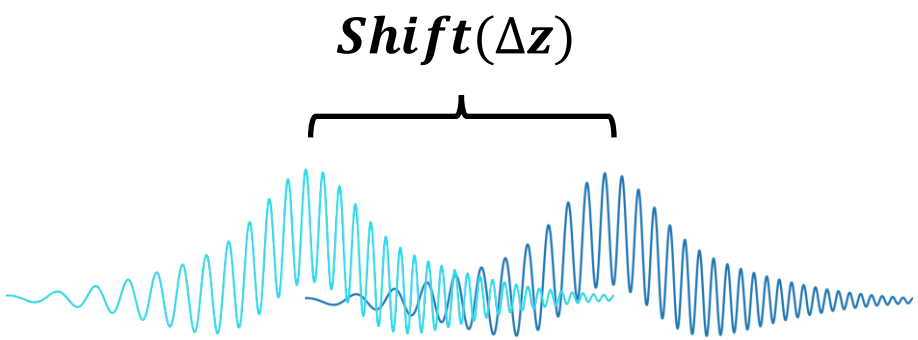


Equation Condition No 1.18



位相限定法によるシフト量



$$F_k(\chi,\eta) = Fourier(f_k(x,y)) \left\{ k = 1,2 \right.$$

$$Shift\ (\Delta x, \Delta y) = argmax\ (\chi,\eta) \left\{ \frac{F_1(\chi,\eta) \cdot \overline{F_2(\chi,\eta)}}{|F_1(\chi,\eta)||F_2(\chi,\eta)|} \right\}$$

記号	意味
$\lvert X \rvert$	X の大きさ
$\bar{X}$	Xの複素共役
$argmax$	最大値となる時の インデックス
$Fourier$	フーリエ変換

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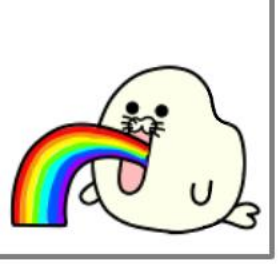
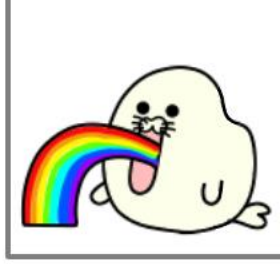
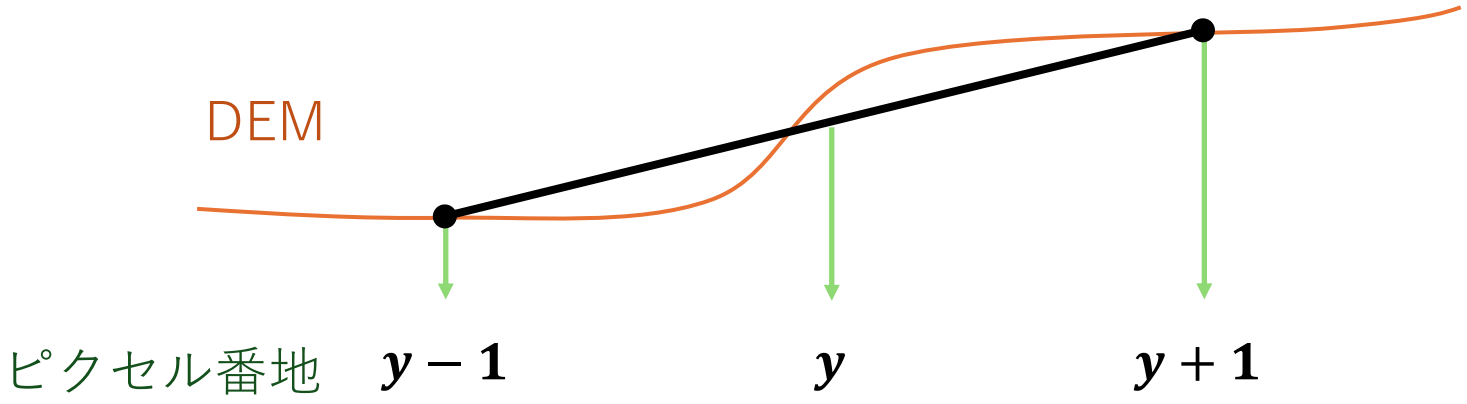


Equation Condition No 1.19



DEMによる傾斜の算出



記号	意味
DEM	デジタル標高モデル(高度)
$Slope$	傾斜

傾斜 $Slope(y) = DEM(y+1) - DEM(y-1)$

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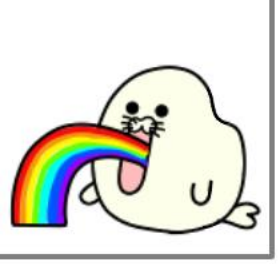
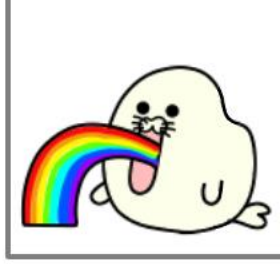
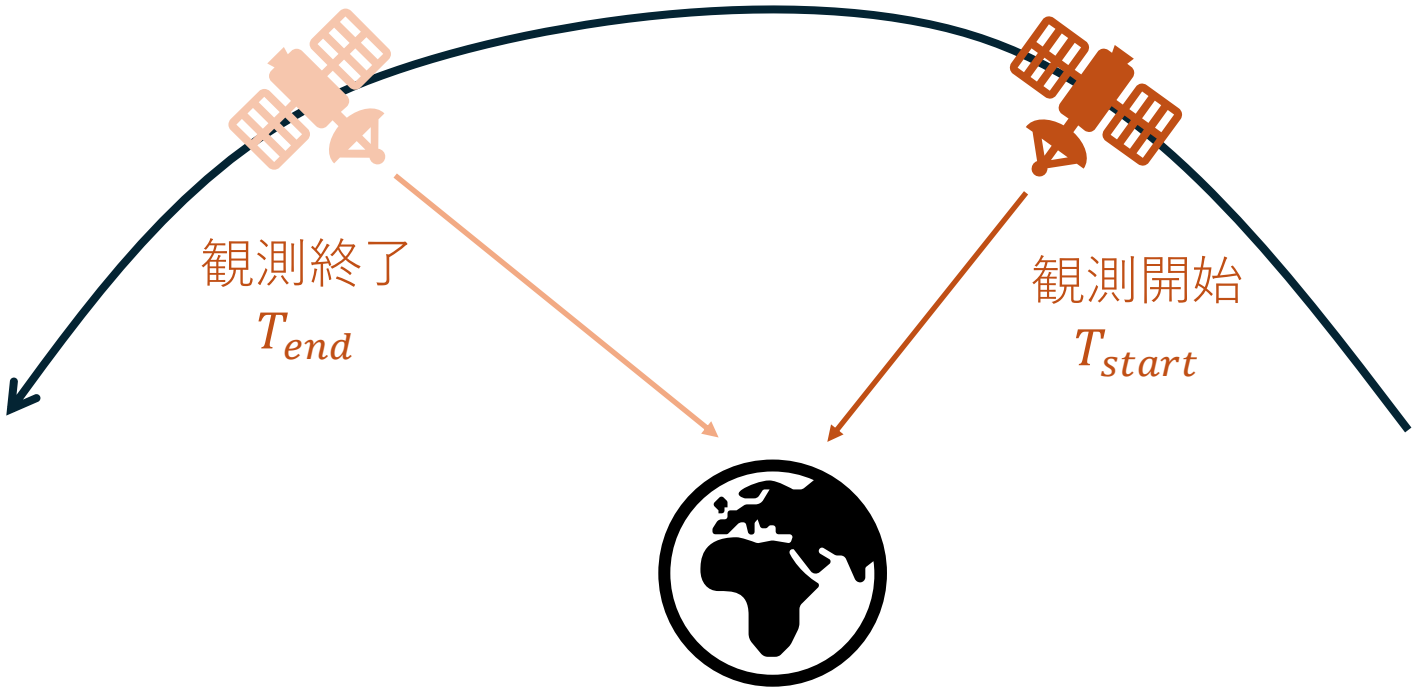


Equation Condition No 2.1



衛星の観測時刻



記号	意味
T_{satrt}	観測開始時刻
T_{end}	観測終了時刻
$T_{observation\ duration}$	観測時間

観測時間 $\mathrm{T}_{Obsevation\ \mathrm{duration}} = T_{end} - T_{start}$

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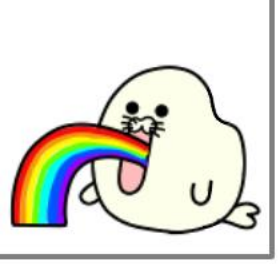
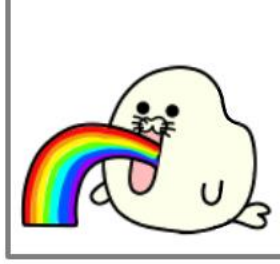
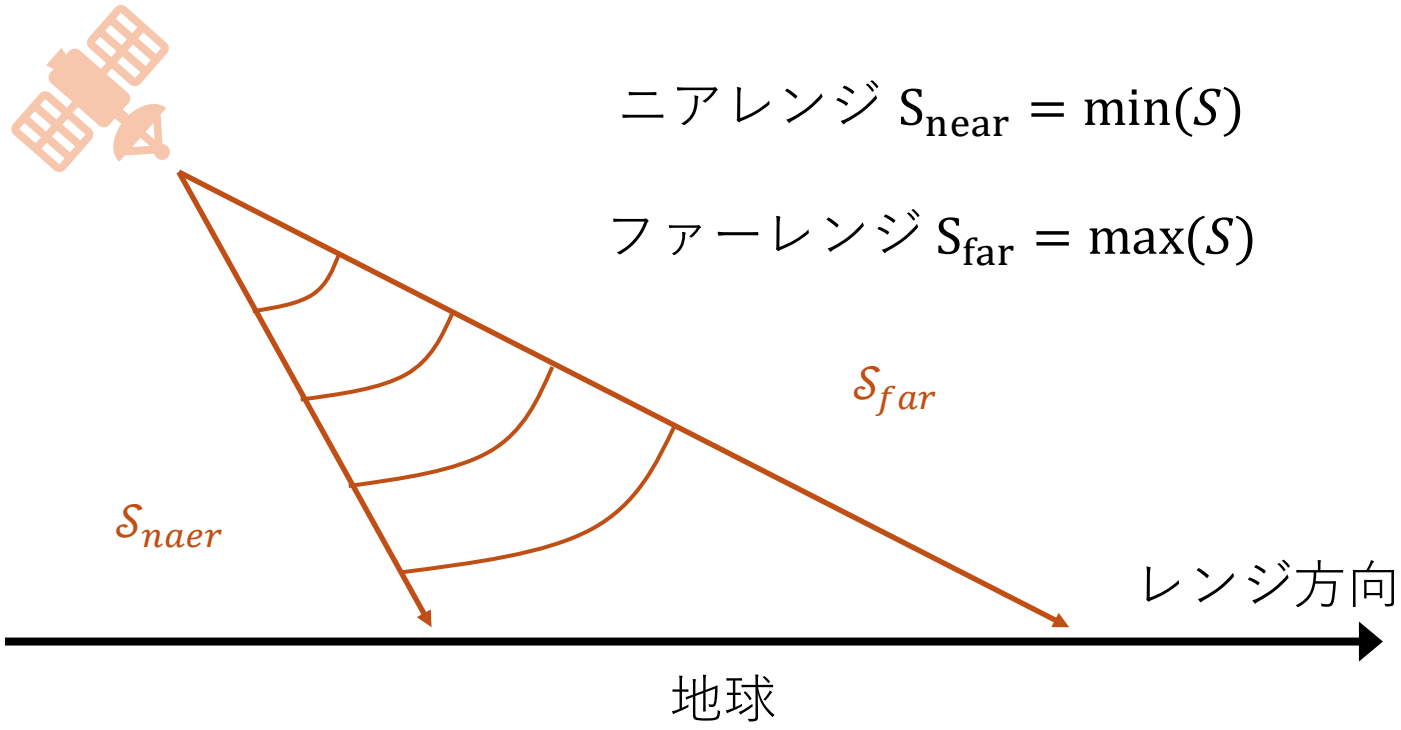


Equation Condition No 2.2





ニアレンジ $S_{near} = \min(S)$

ファーレンジ $S_{far} = \max(S)$

記号	意味
S	スラントレンジ
S_{near}	ニアレンジ
S_{far}	ファーレンジ

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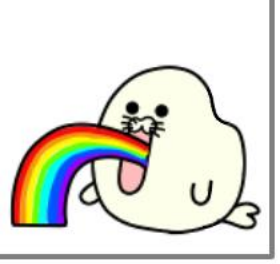
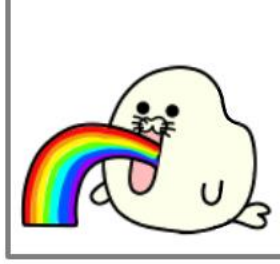
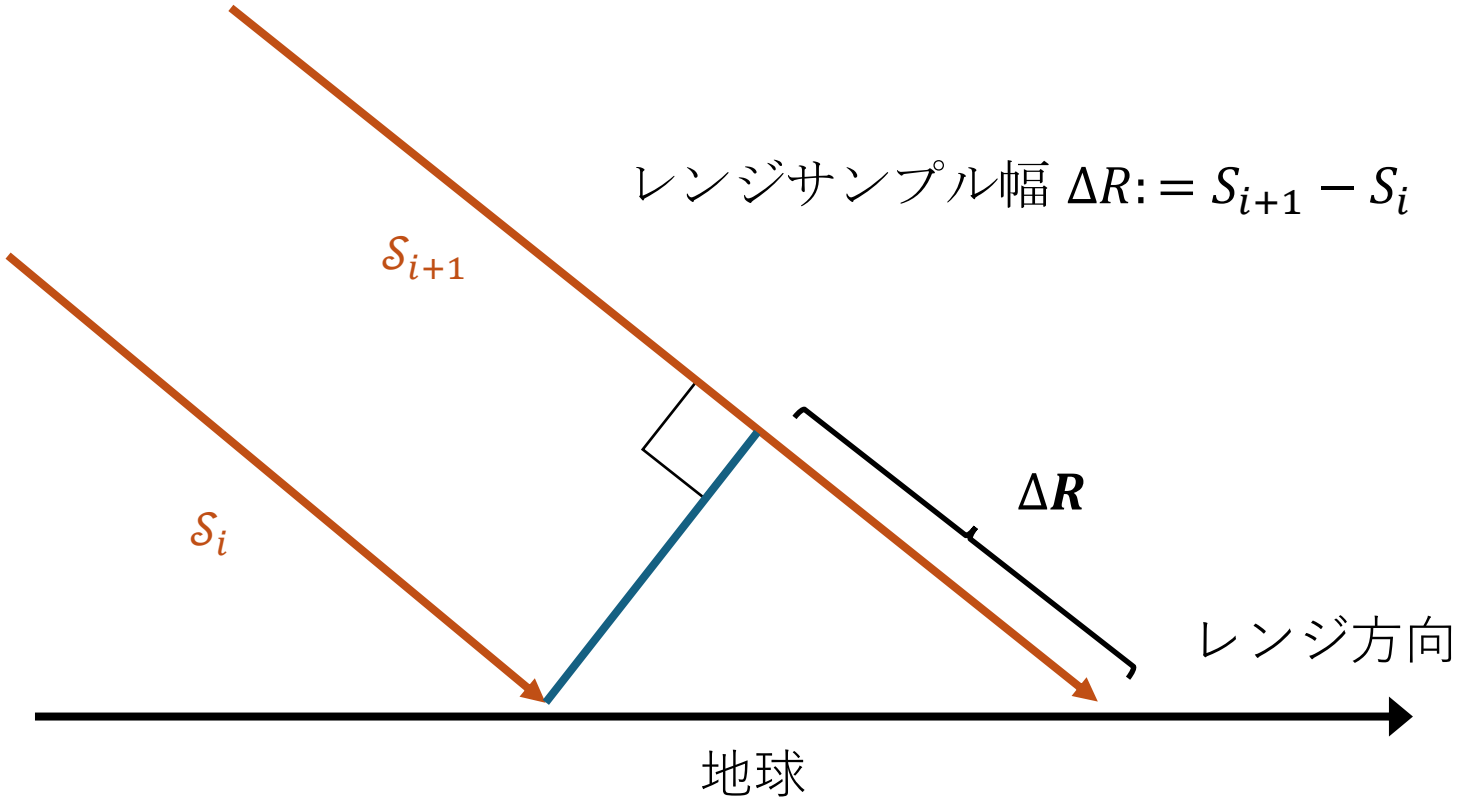


Equation Condition No 2.3





記号	意味
S_i	i番目の時のスラントレンジ
ΔR	レンジサンプル幅 スラントレンジ解像度

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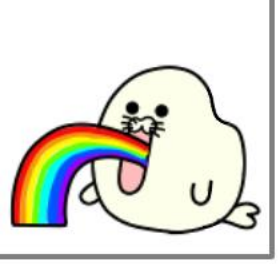
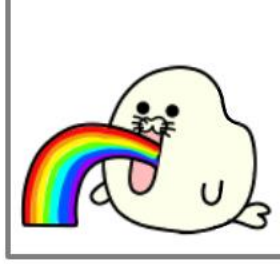
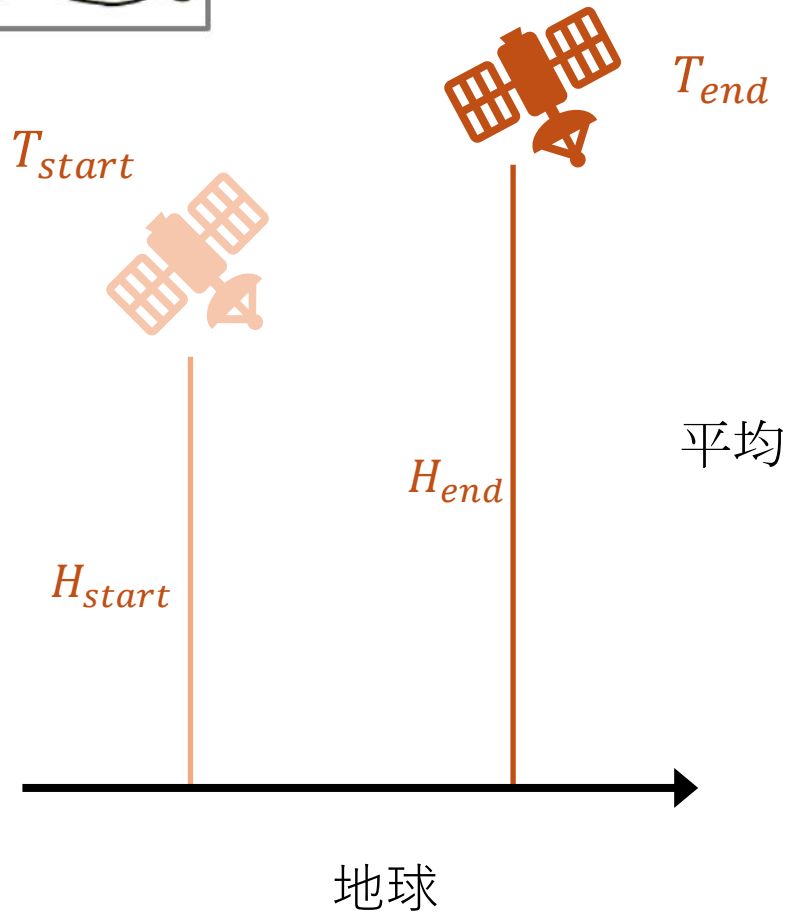


Equation Condition No 2.4





$$\text{平均高度 } \mathrm{H_{mean}} = \int_{T_{sart}}^{T_{end}} H(T_i)\, dT$$

記号	意味
H_{mean}	平均衛星高度
H_{start}	観測開始時の衛星高度
H_{end}	観測終了時の衛星高度
T_i	サンプルi番目の時刻

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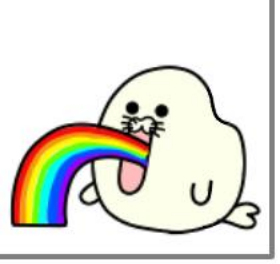
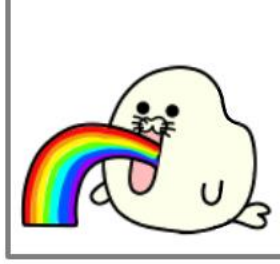
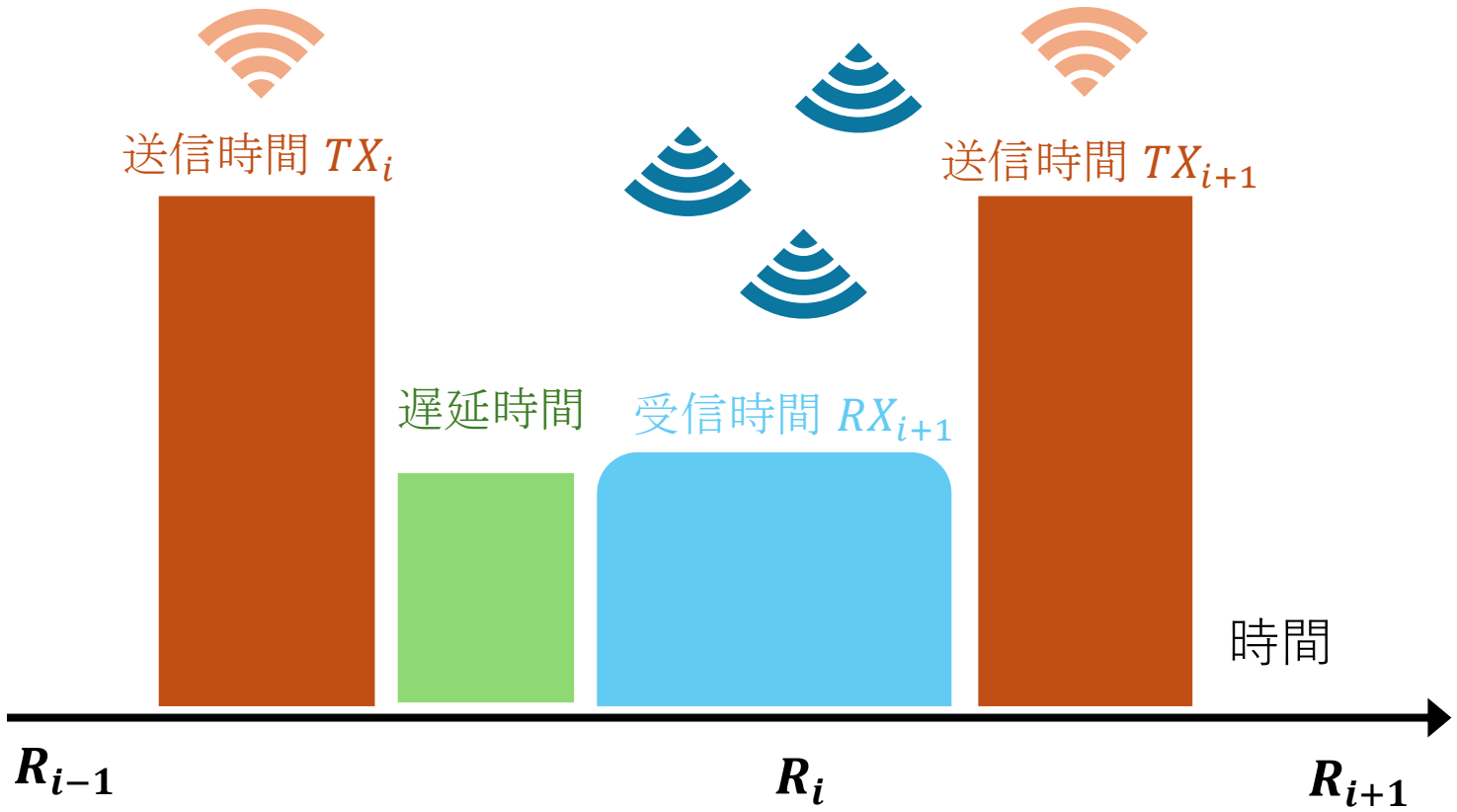


Equation Condition No 2.5





記号	意味
TX_i	i番目の送信時間
RX_i	i番目の受信時間
T_{offset}	時間シフト

$$Time\ Gate\ Delay = T_{RX_i} - T_{offset}$$

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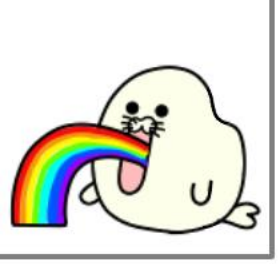
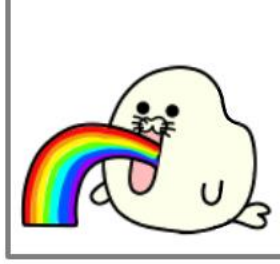


Equation Condition No 2.6,7,8,9,10



時間の単位を Sec (Hz), m に変換

$$Sampling\ Frequency = f_s[MHz] * 10^6$$

$$PRF = PRF[mHz] * 10^{-3}$$

$$Pulse\ Duration = T_{chirp}[nsec] * 10^{-9}$$

$$地球半径 = ellipsoid\ radius[km] * 10^3$$

$$地球短半径 = ellipsoid\ short\ radius[km] * 10^3$$

記号	意味
PRF	Pulse Repetition Frequency パルス繰り返し周波数
f_s	ADサンプリング周波数 レンジサンプリング数
T_{chirp}	チャープ信号の送信時間

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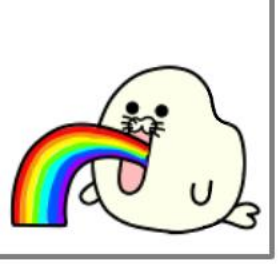
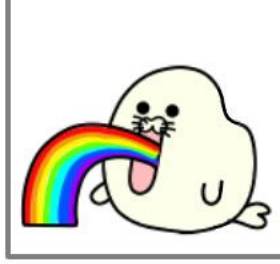
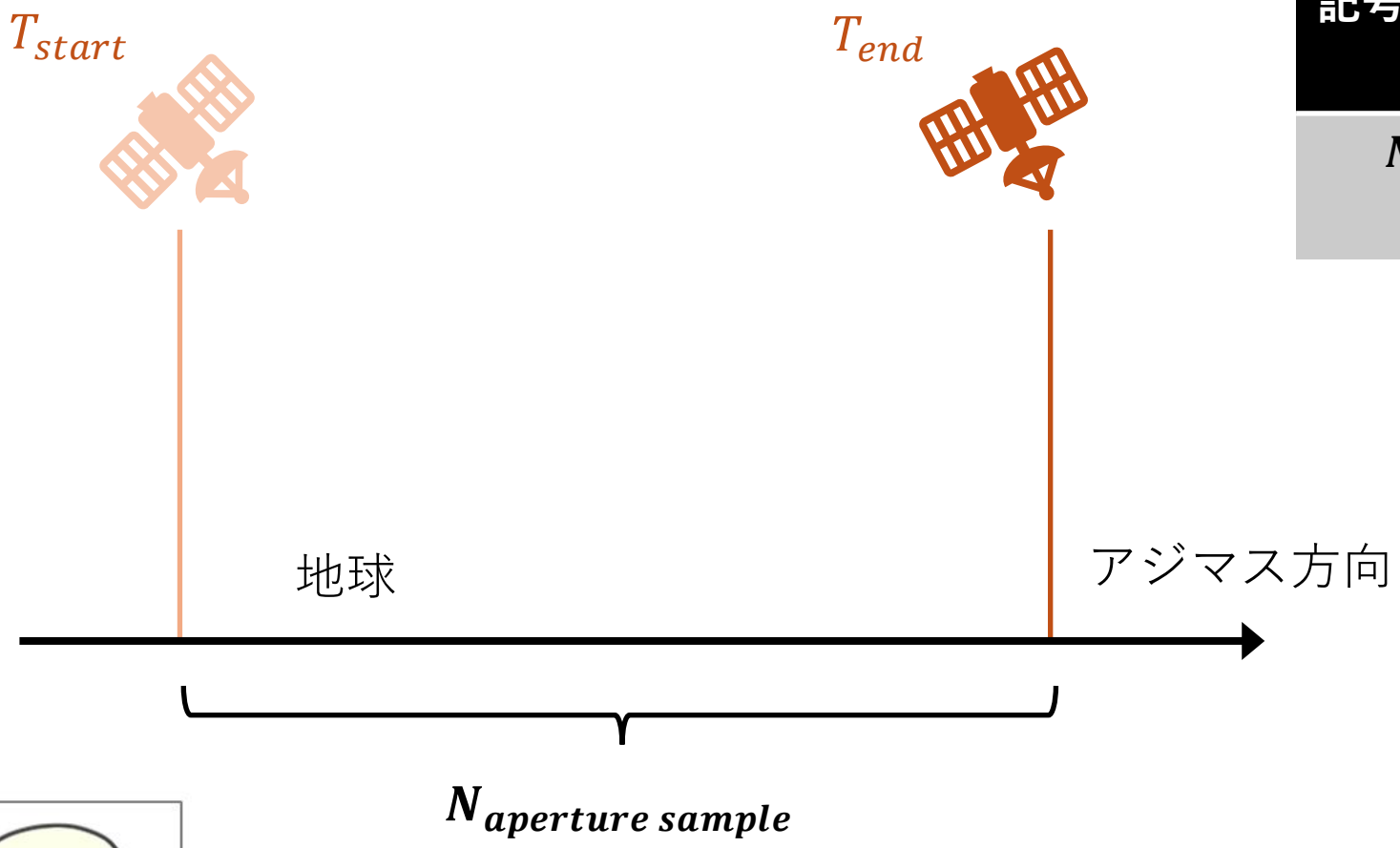


Equation Condition No 2.11





記号	意味
$N_{aperture\ sample}$	合成開口数 アジマスサンプル数

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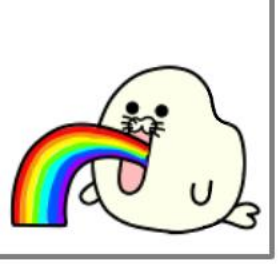
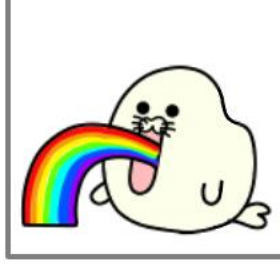
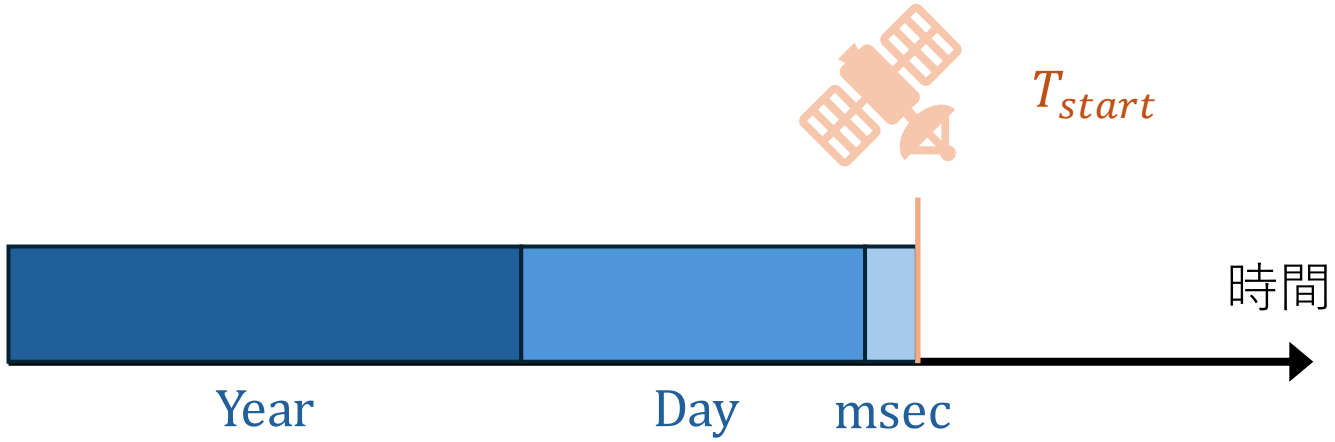


Equation Condition No 2.12



観測開始時刻を秒単位に変換



記号	意味
T_{shift}	時刻誤差
T_{day}	観測時刻の日付

$$T_{Observation\ day\ fraction} = Day + \frac{\frac{msec}{10^4} + T_{shift}}{T_{day}}$$

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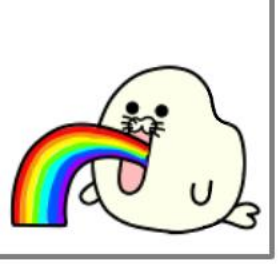
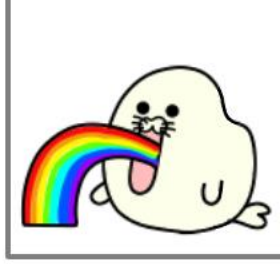
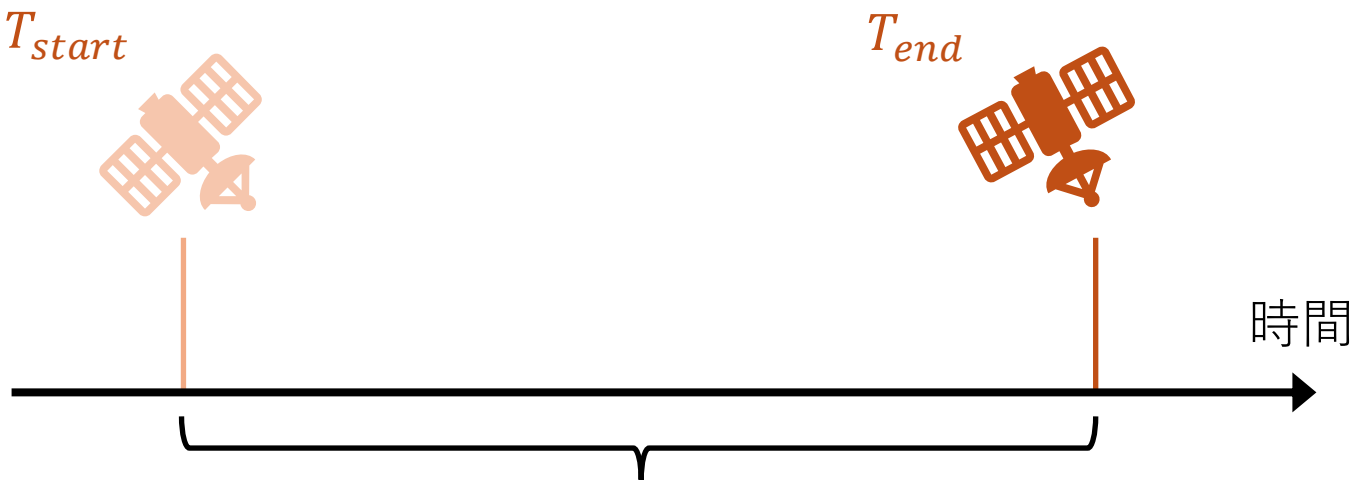


Equation Condition No 2.13,14





$$T_{observation} = N_{aperture\ sample} * PRF$$

記号	意味
$T_{observation\ day\ fraction}$	1日あたりの秒数
$N_0 = 0$	サンプルなし
$N_{aperture\ sample}$	合成開口数 アジマスサンプル数
PRF	パルス繰り返し周波数

$$T_{start} = T_{Observation\ day\ fraction} * Day + N_0 * PRF$$

$$T_{end} = T_{Observation\ day\ fraction} * Day + N_{aperture\ sample} * PRF$$

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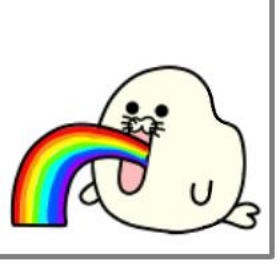
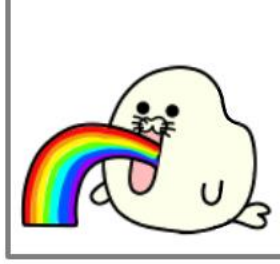
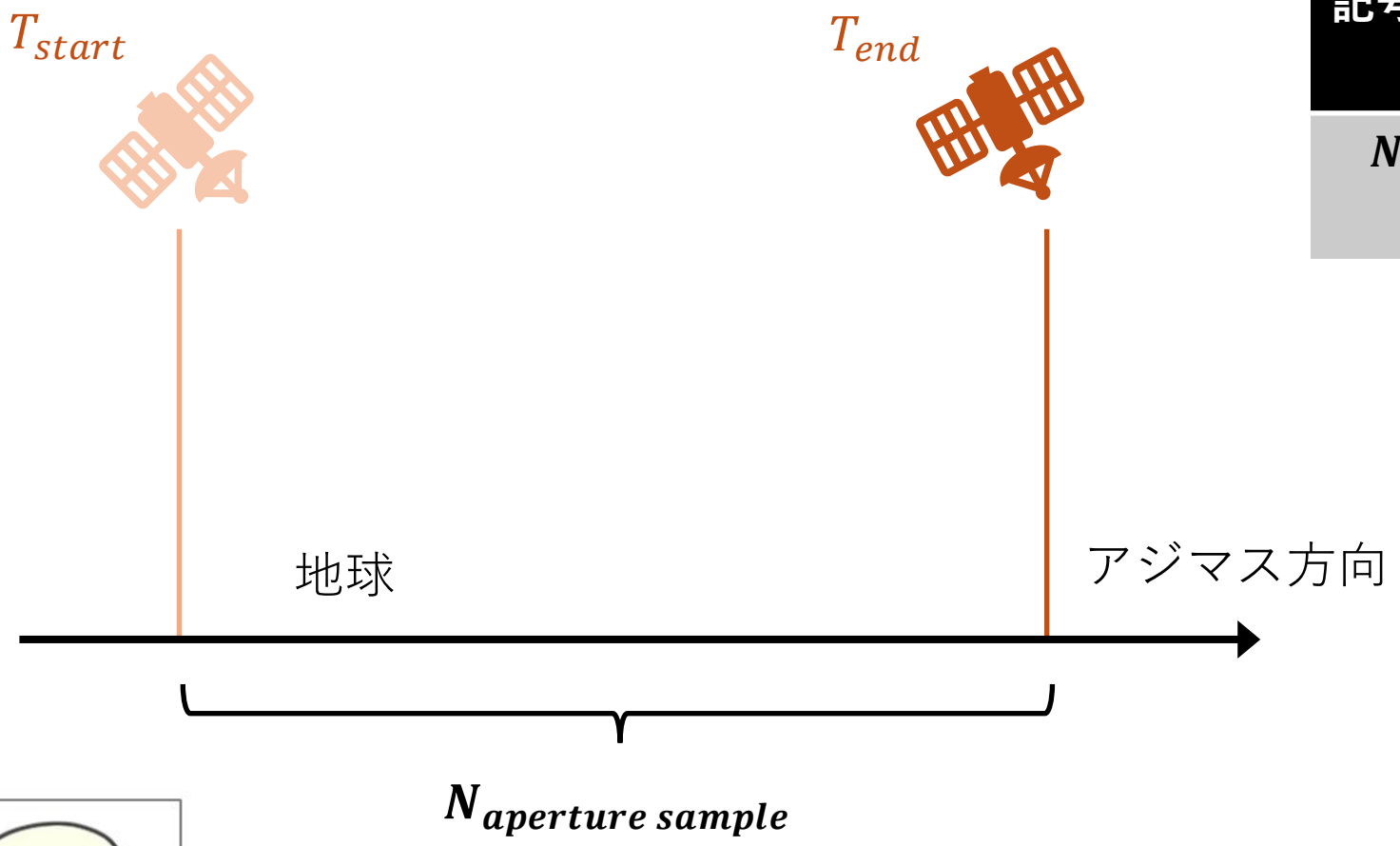


Equation Condition No 2.15





記号	意味
$N_{aperture\ sample}$	合成開口数 アジマスサンプル数

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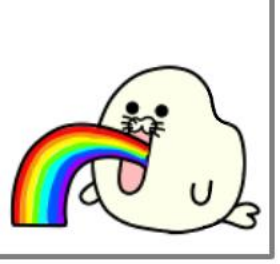
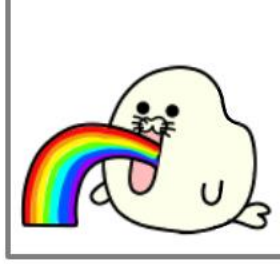
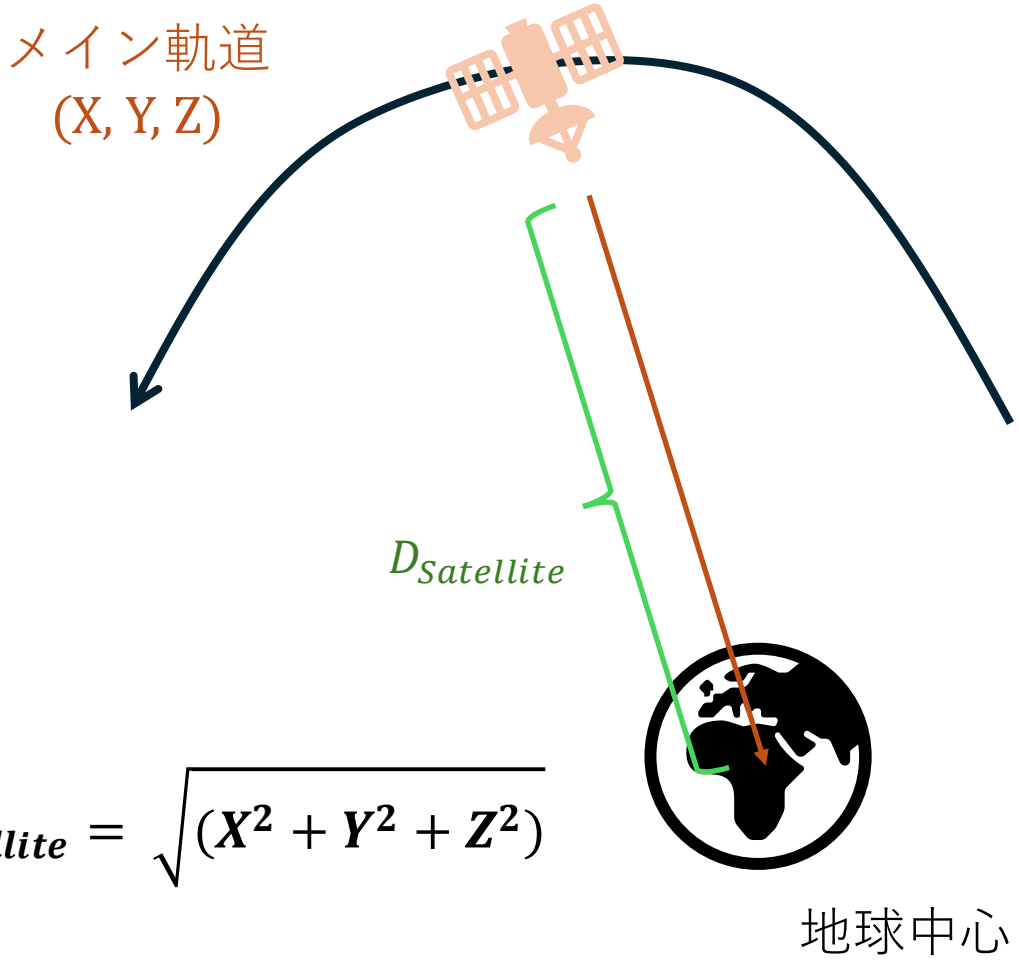


Equation Condition No 2.16





記号	意味
$D_{satellite}$	衛星の地球中心からの距離
X	ECEF座標のx軸成分
Y	ECEF座標のy軸成分
Z	ECEF座標のz軸成分

衛星距離 $D_{satellite} = \sqrt{(X^2 + Y^2 + Z^2)}$

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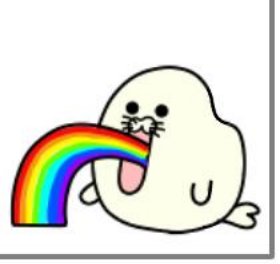
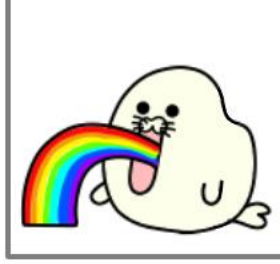
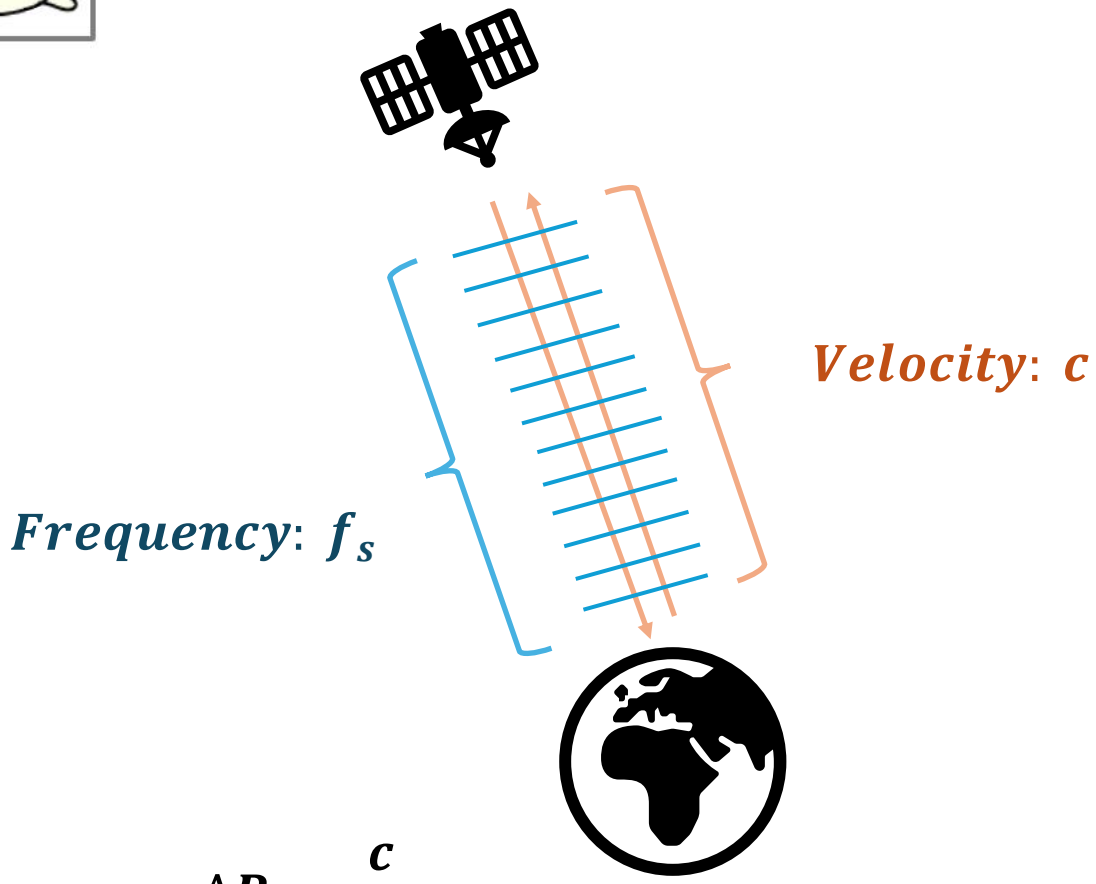


Equation Condition No 2.17





$$\Delta R = \frac{c}{2f_s}$$

記号	意味
c	光速(一定)
f_s	ADサンプリング周波数 レンジサンプリング数
ΔR	レンジサンプル幅 スラントレンジ解像度

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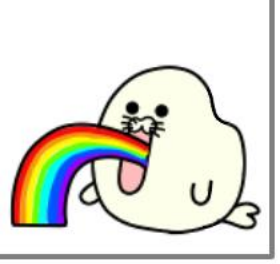
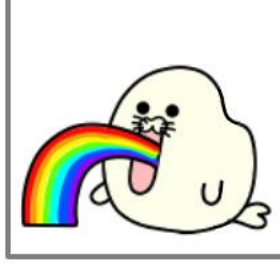
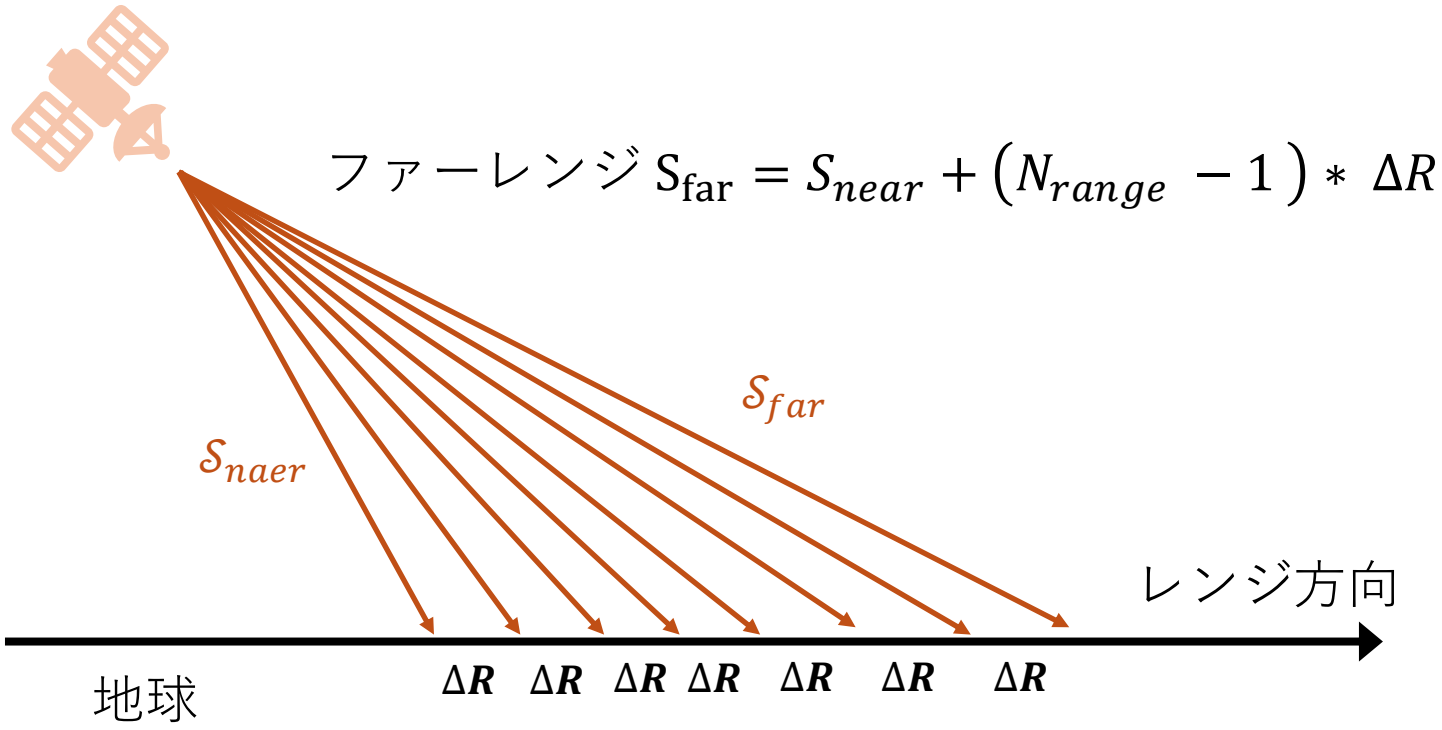


Equation Condition No 2.18,19





記号	意味
N_{range}	レンジサンプル数
S_{near}	ニアレンジ
S_{far}	ファーレンジ
ΔR	レンジサンプル幅 スラントレンジ解像度

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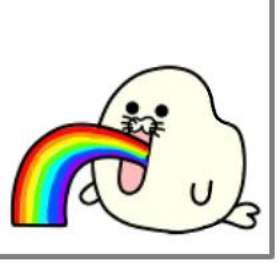
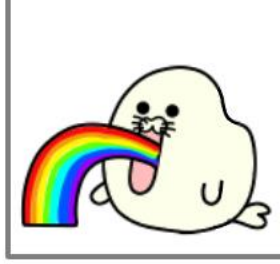
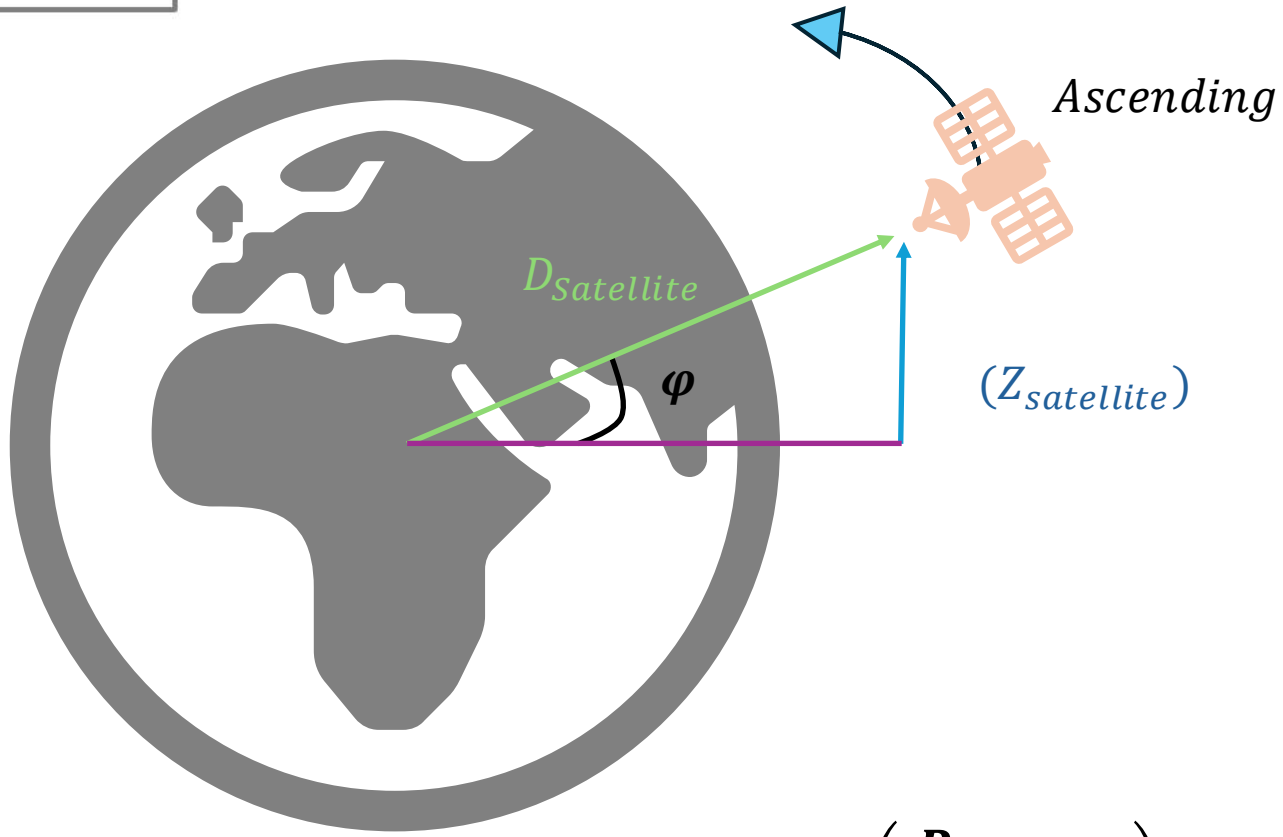


Equation Condition No 2.20





記号	意味
$D_{Satellite}$	衛星の地球中心からの距離
$Z_{satellite}$	衛星のECEF座標のZ軸の変化
$P_{satellite}$	衛星の位置座標
φ	衛星位置の緯度(角度)

$$\varphi = ArcTan\left(\frac{P_{satellite}}{Z_{psatellite}}\right)$$

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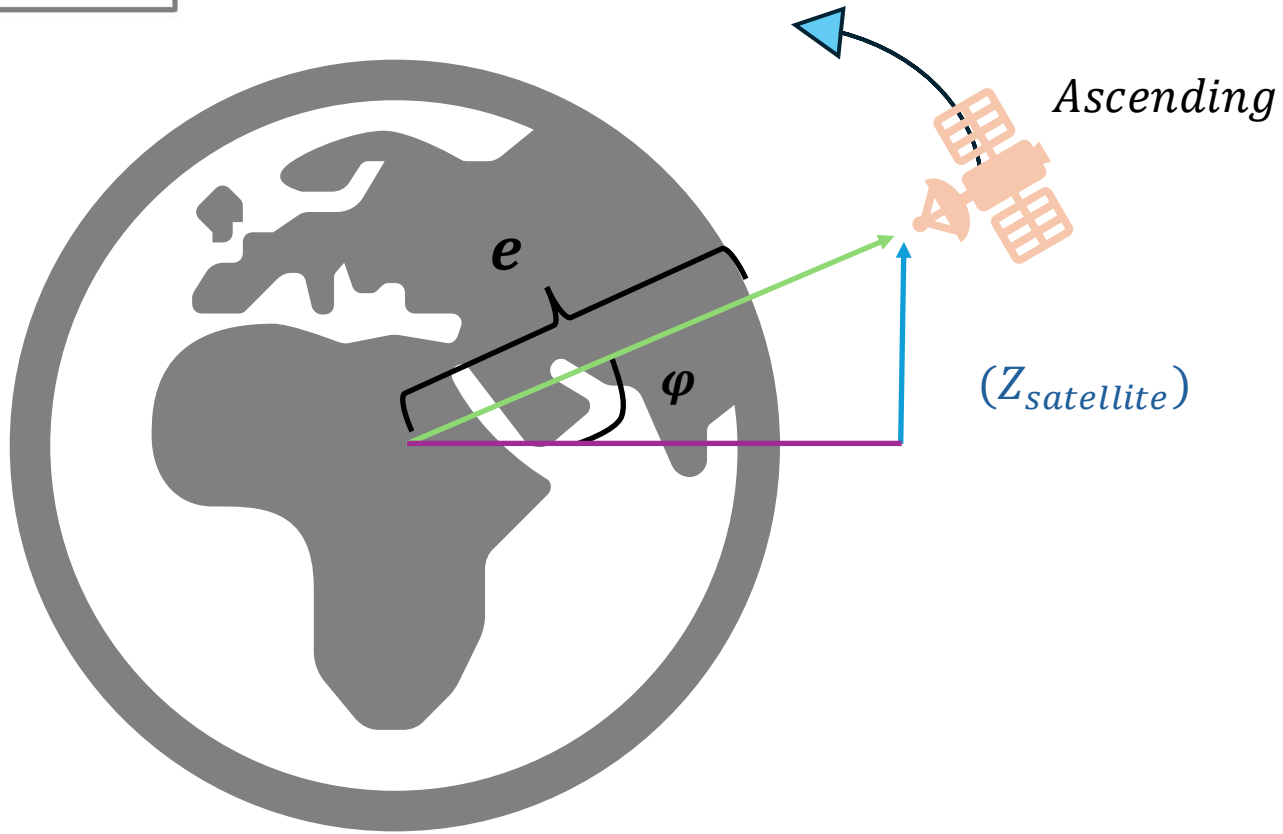
Equation Condition No 2.21



記号	意味
e	衛星位置での地球半径
a	長半径(赤道半径)
b	短半径(極半径)
φ	衛星位置の緯度(角度)

$$Earth\ Radius:\ e = \frac{1}{\sqrt{\left(\frac{cos^2(\varphi)}{a^2}\right) + \left(\frac{sin^2(\varphi)}{b^2}\right)}}$$

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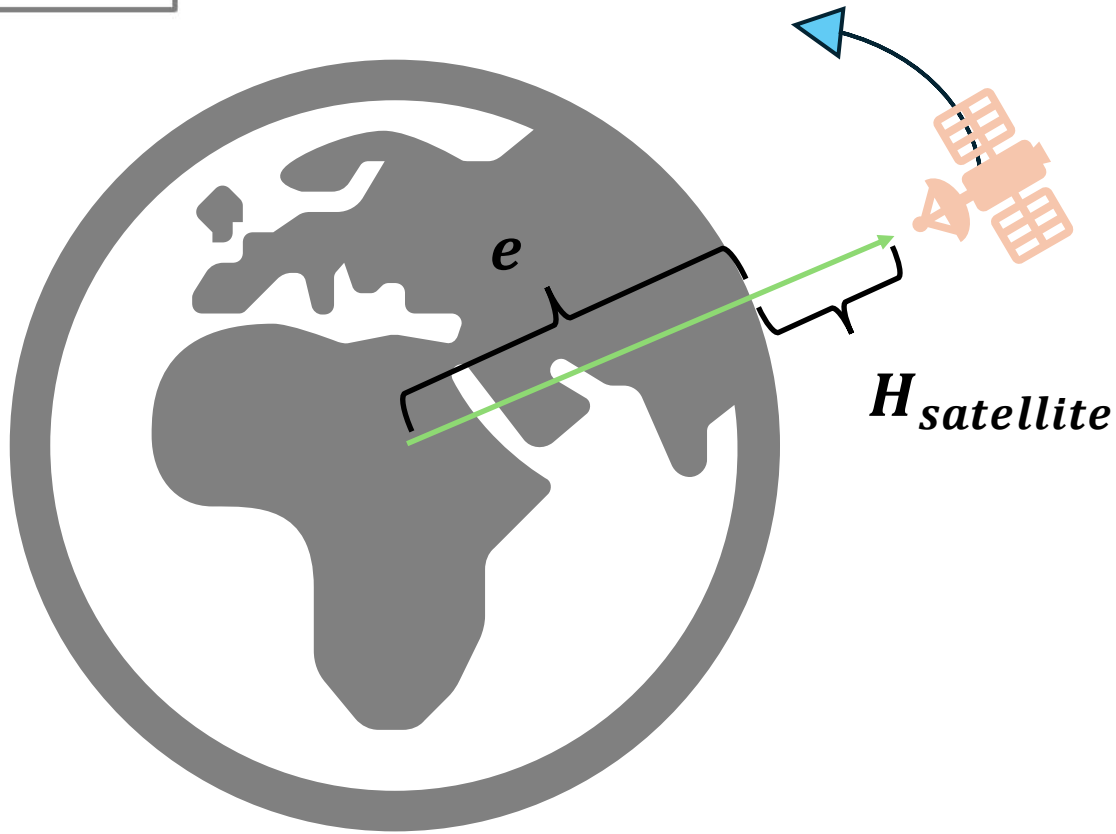
Equation Condition No 2.22



記号	意味
e	衛星位置での地球半径
$H_{satellite}$	衛星の高度
$D_{Satellite}$	衛星の地球中心からの距離

衛星高度 $H_{satellite} = D_{satellite} - e$

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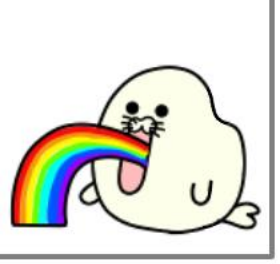
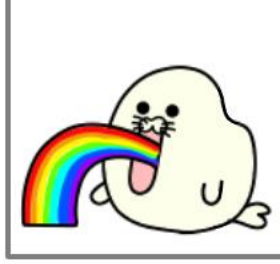
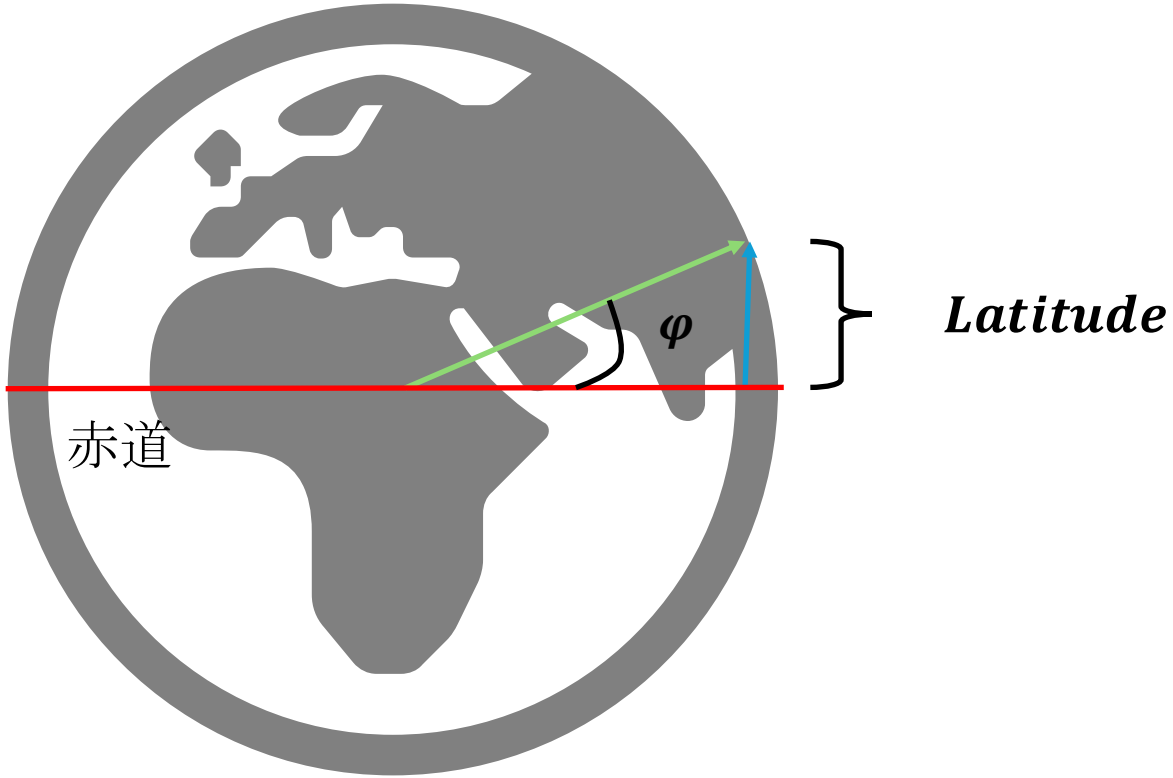


Equation Condition No 3.1





記号	意味
φ	緯度(角度)
f	地球楕円体の扁平率

$$Latitude = ArcTan((1 - f)^2 * Tan(\varphi))$$

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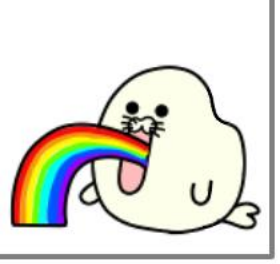
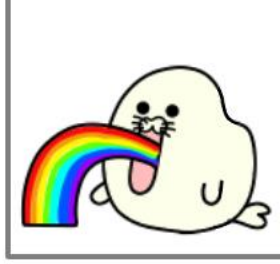
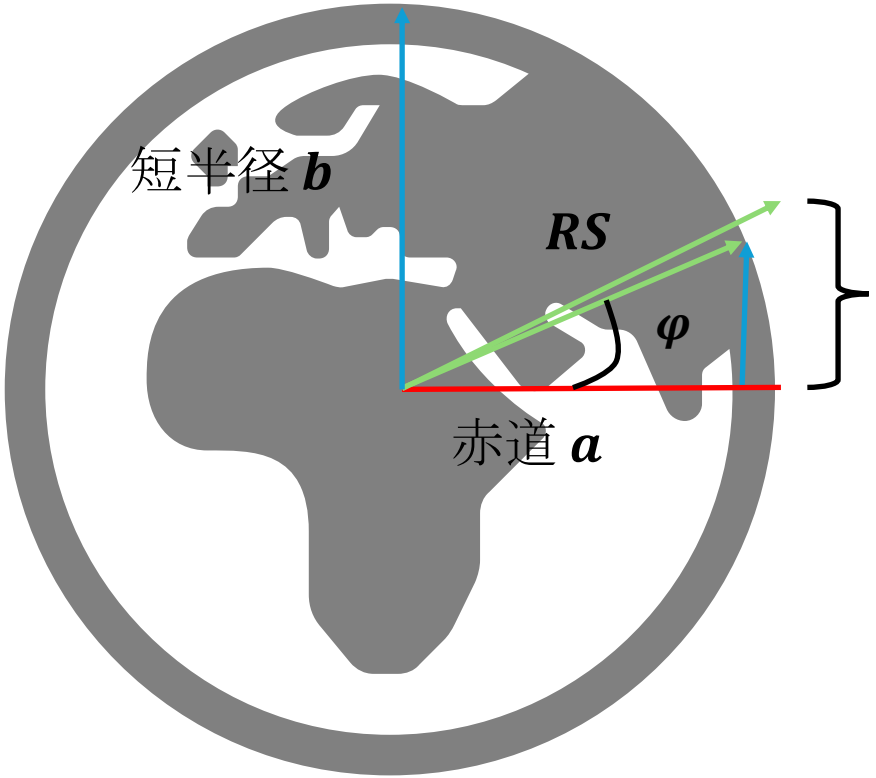


Equation Condition No 3.2





$Latitude_{+\text{height}}$: φc

記号	意味
a	長半径(赤道半径)
b	短半径(極半径)
φ	観測地点の緯度(角度)
φc	観測地点と地形を含めた緯度(角度)

$$Ellipsoid\ Radius = \frac{a}{\sqrt{1 + cos^2(\varphi c)((1 - f)^2 - 1)}}$$

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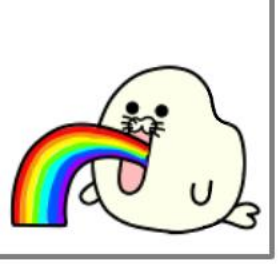
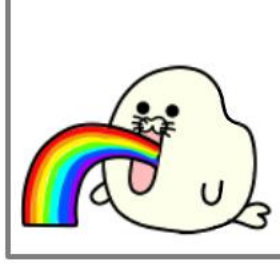
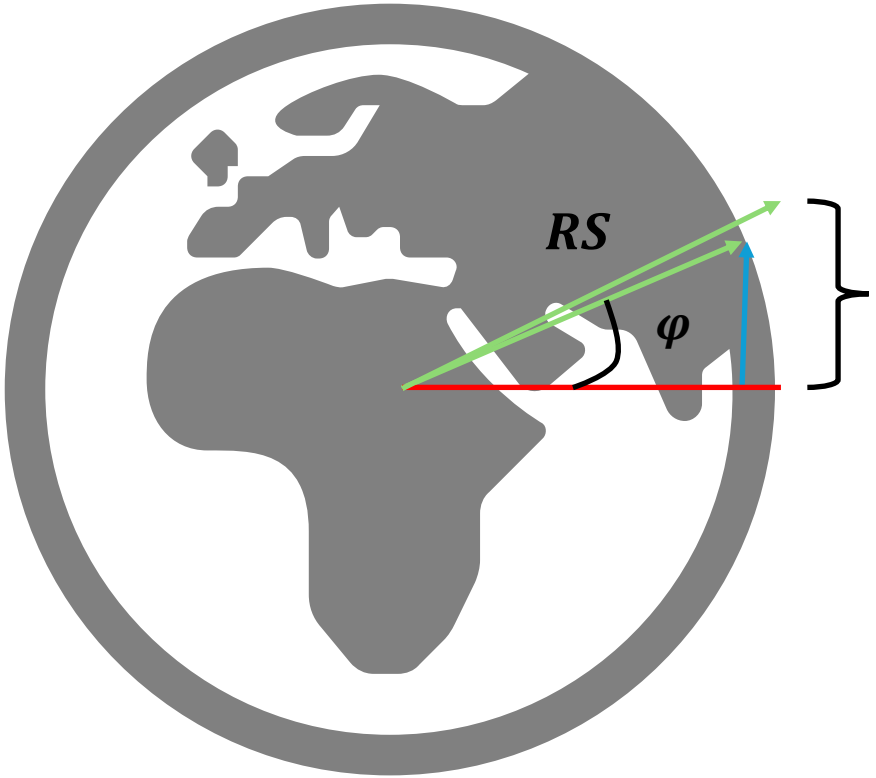


Equation Condition No 3.3





$Latitude_{+\mathrm{height}}$: φc

記号	意味
φ	観測地点の緯度(角度)
φc	観測地点と地形を含めた緯度(角度)
RS	φcの時の地球中心からの距離
R	φcの時の地球半径

$$Geodetic\ Distance: R = \sqrt{H^2 + rR + 2(H * RS)cos(\varphi - \varphi c)}$$

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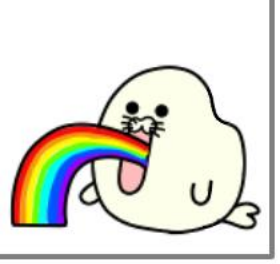
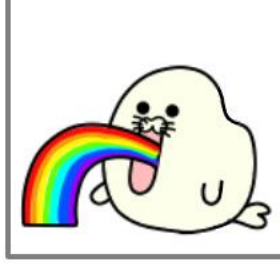
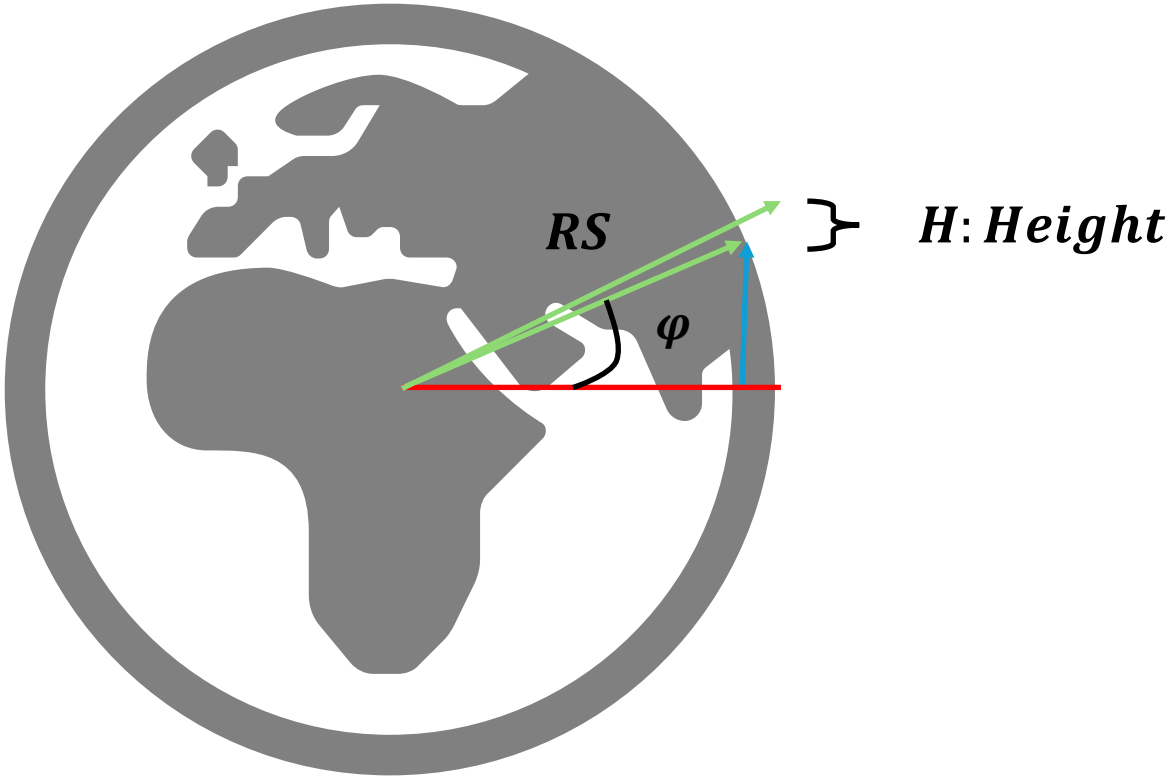


Equation Condition No 3.4





記号	意味
φ	観測地点の緯度(角度)
φc	観測地点と地形を含めた緯度(角度)
H	高さ(地形)
R	φcの時の地球半径

$$Geodetic\ Latitude = \varphi c + ArcSin\left(\frac{H * sin(\varphi - \varphi c)}{R}\right)$$

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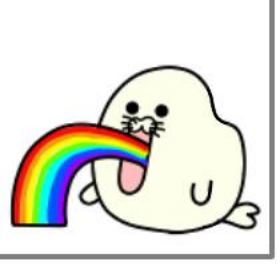
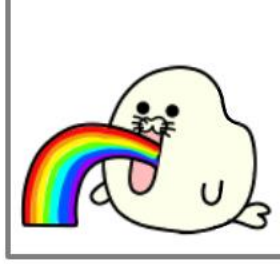
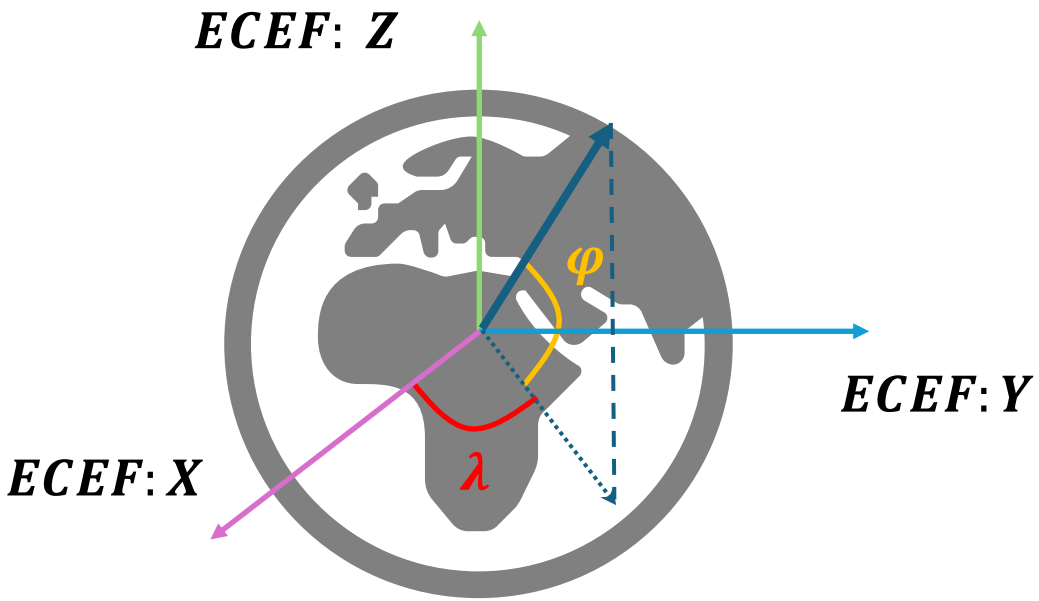


Equation Condition No 3.5





記号	意味
(X, Y, Z)	ECEF座標の3軸
φ	緯度(角度)
λ	経度(角度)
R	φcの時の地球半径

$$x = R * cos(\varphi) * cos(\lambda)$$
$$y = R * cos(\varphi) * sin(\lambda)$$
$$z = R * sin(\varphi)$$

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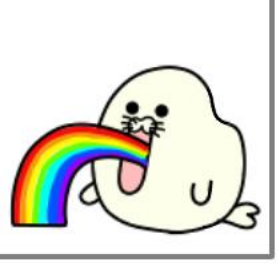
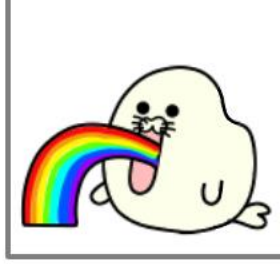
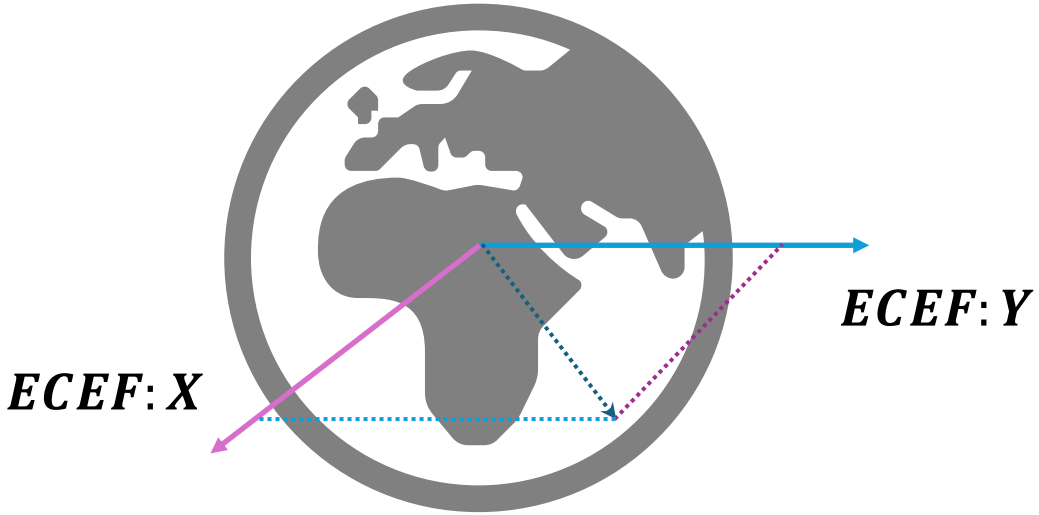


Equation Condition No 3.7





平面距離 $p = \sqrt{x^2 + y^2}$

記号	意味
(X, Y, Z)	ECEF座標の3軸
φ	緯度(角度)
λ	経度(角度)
R	φcの時の地球半径

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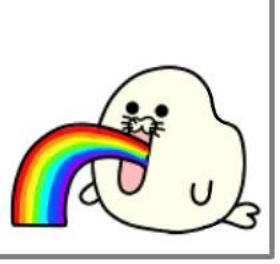
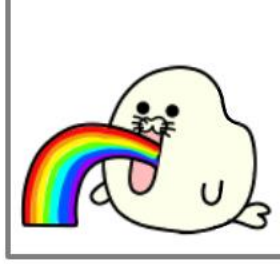
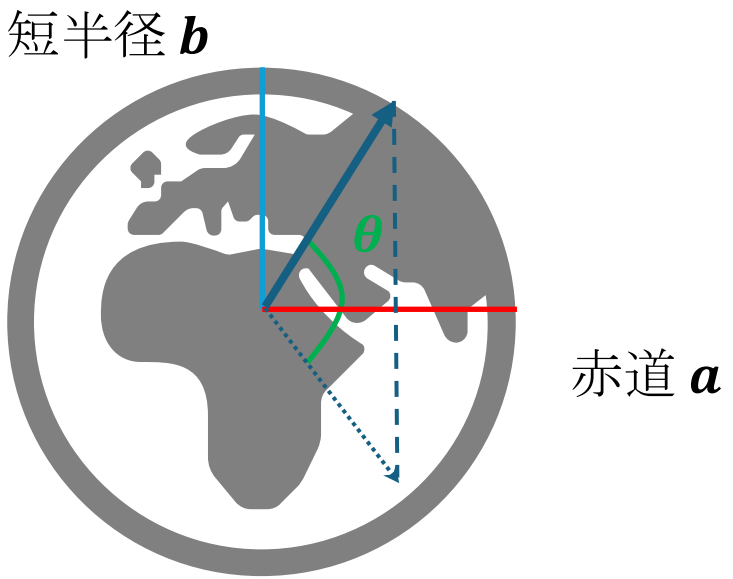


Equation Condition No 3.8





$$\theta = ArcTan\left(\frac{Z * a}{(p * b)}\right)$$

記号	意味
Z	ECEF座標のZ軸
p	XY平面上の距離
a	長半径(赤道半径)
b	短半径(極半径)

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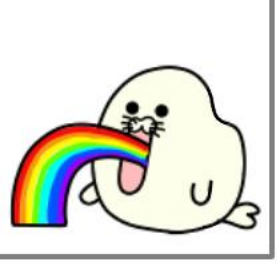
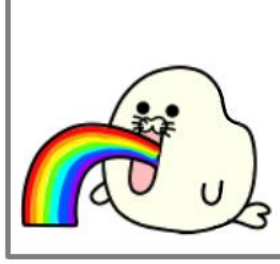
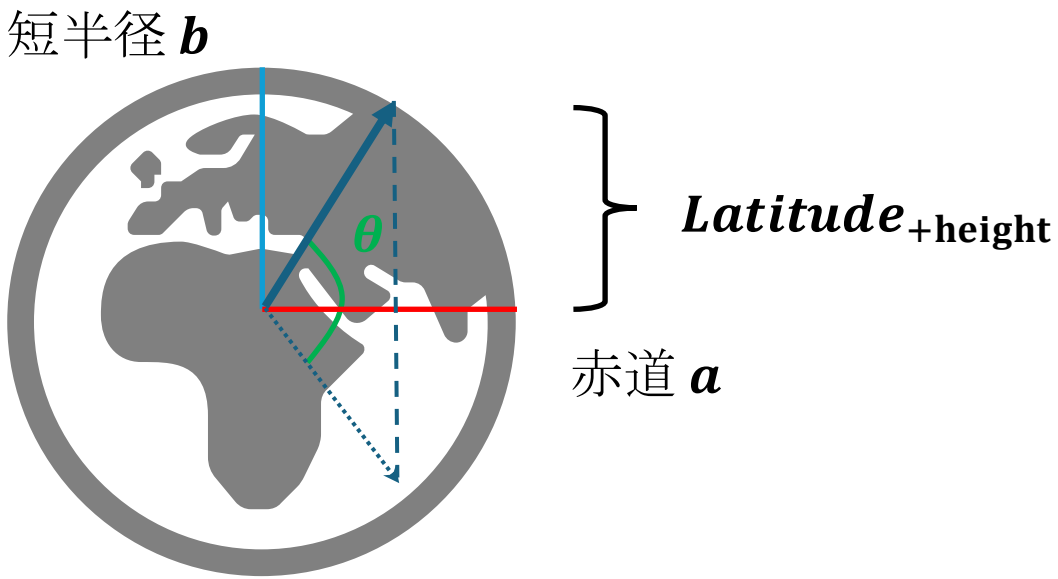


Equation Condition No 3.9





記号	意味
Z	ECEF座標のZ軸
p	XY平面上の距離
a	長半径(赤道半径)
b	短半径(極半径)
θ	XY平面投影の緯度(角度)
f	地球楕円体の扁平率

$$Geodetec\ Latitude = ArcTan\left(\frac{Z + f * b * sin^3(\theta)}{p - f * a * cos^3(\theta)}\right)$$

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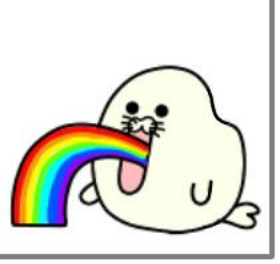
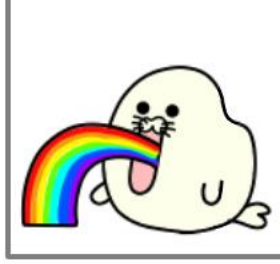
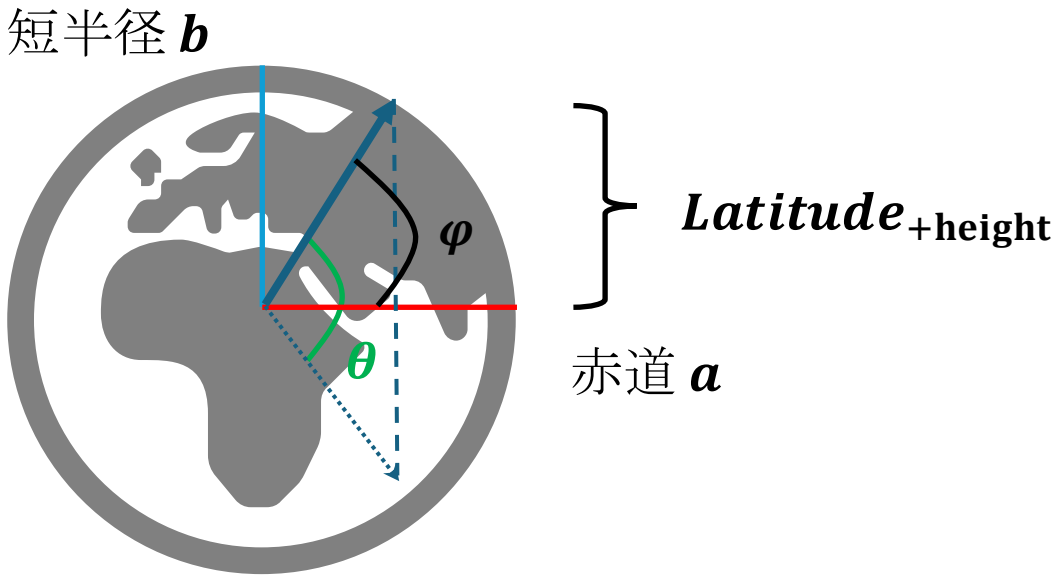


Equation Condition No 3.10





$$Curvature: N = \frac{a}{\sqrt{(1 - e(2 - e)sin^2(\varphi)}}$$

記号	意味
Z	ECEF座標のZ軸
p	XY平面上の距離
a	長半径(赤道半径)
b	短半径(極半径)
φ	緯度(角度)
e	地球半径

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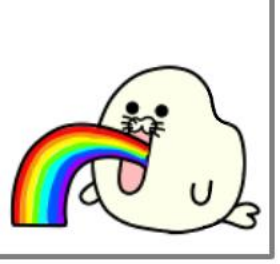
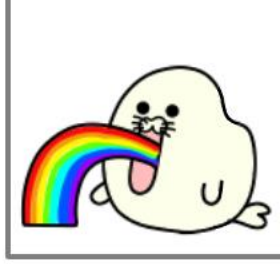
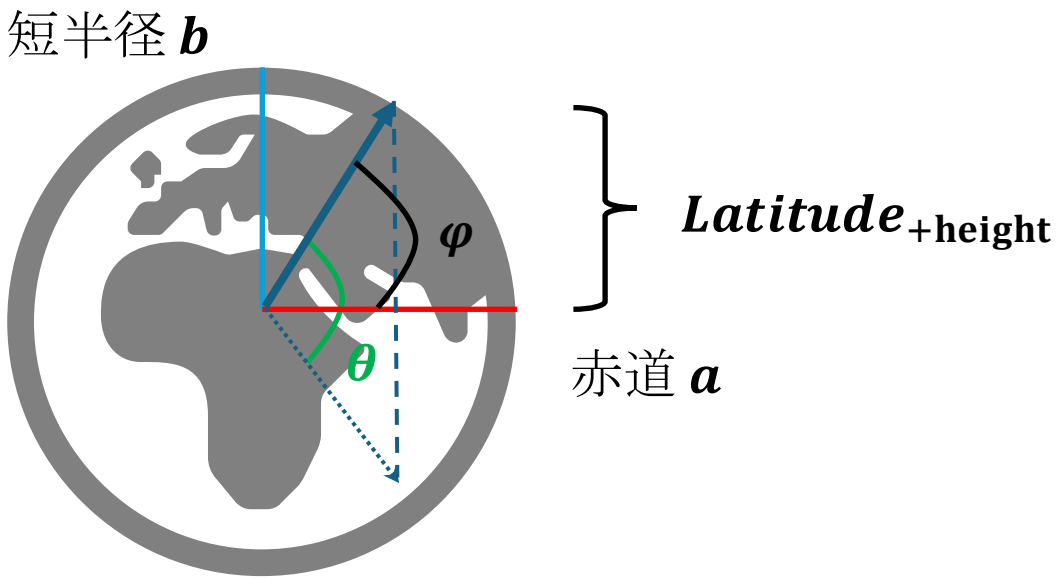


Equation Condition No 3.11





記号	意味
z	ECEF座標のZ軸
p	XY平面上の距離
a	長半径(赤道半径)
b	短半径(極半径)
φ	緯度(角度)
e	地球半径

$$Height := \begin{cases} \dfrac{p}{cos(\varphi)} - N, & |Latitude| < \dfrac{\pi}{4} \\ \dfrac{z}{sin(\varphi)} - N\,(1 - (2 - e)e), & |Latitude| \geq \dfrac{\pi}{4} \end{cases}$$

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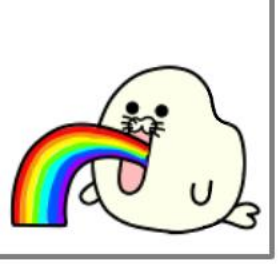
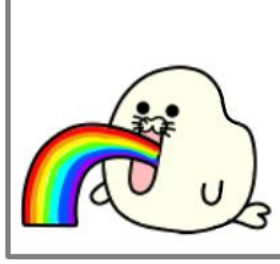
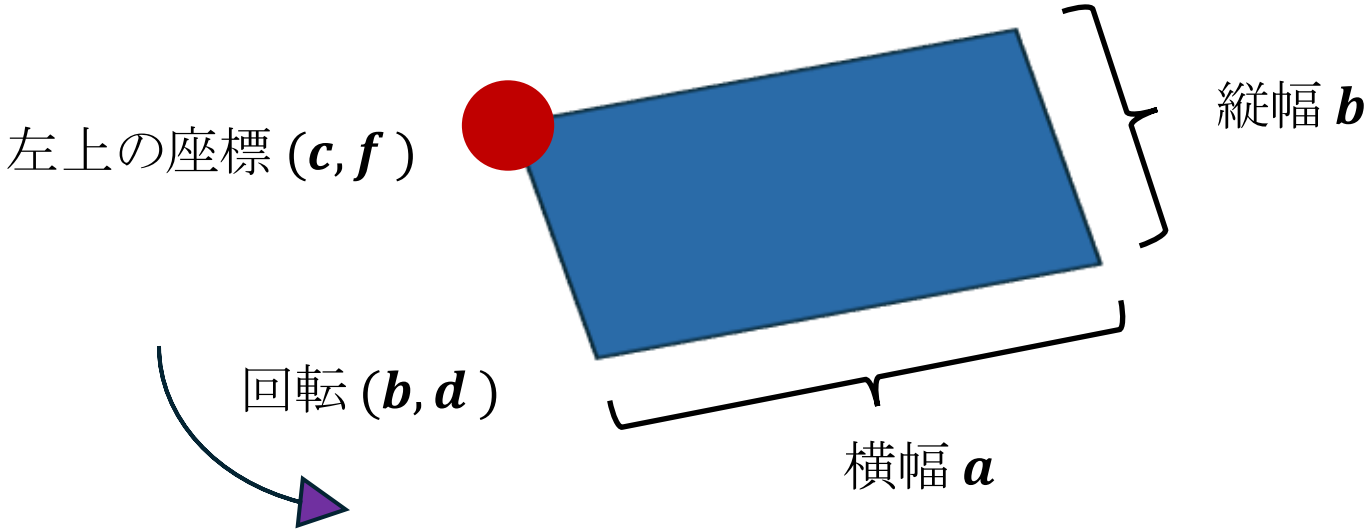


Equation Condition No 3.12



GeoTIFF のTransfrom行列



$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} := \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i \\ j \\ 1 \end{bmatrix}$$

$$x_i = ai + bj + c$$

$$y_i = di + ej + f$$

記号	意味
x_i	i番目の地理空間座標
y_j	j番目の地理空間座標
i	横幅のピクセルインデックス
j	縦幅のピクセルインデックス

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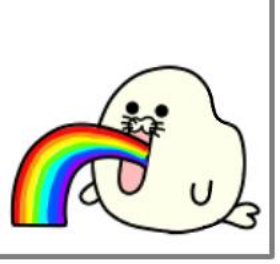
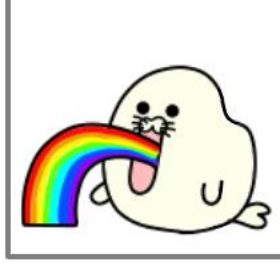
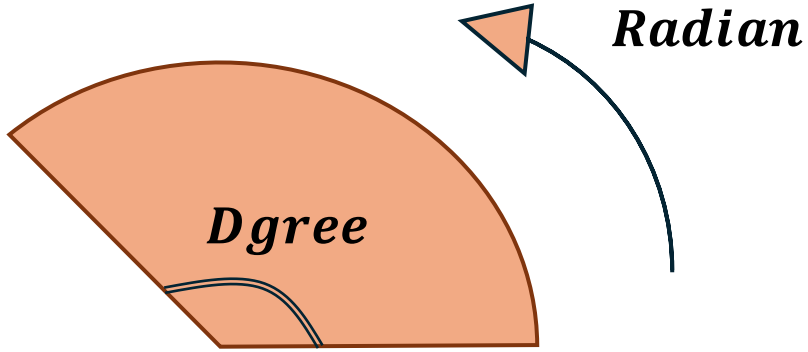


Equation Condition No 3.13



角度のラジアン変換



$$Radian := \frac{deg}{180}\pi$$

記号	意味
π	円周率
$Radian$	円の弧の長さ
$Dgree$	角度

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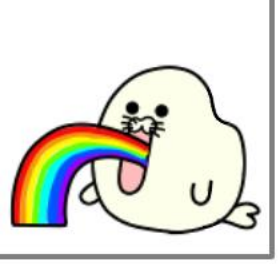
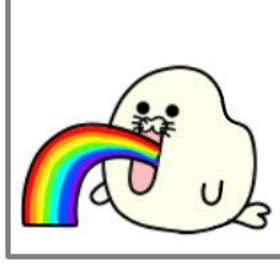
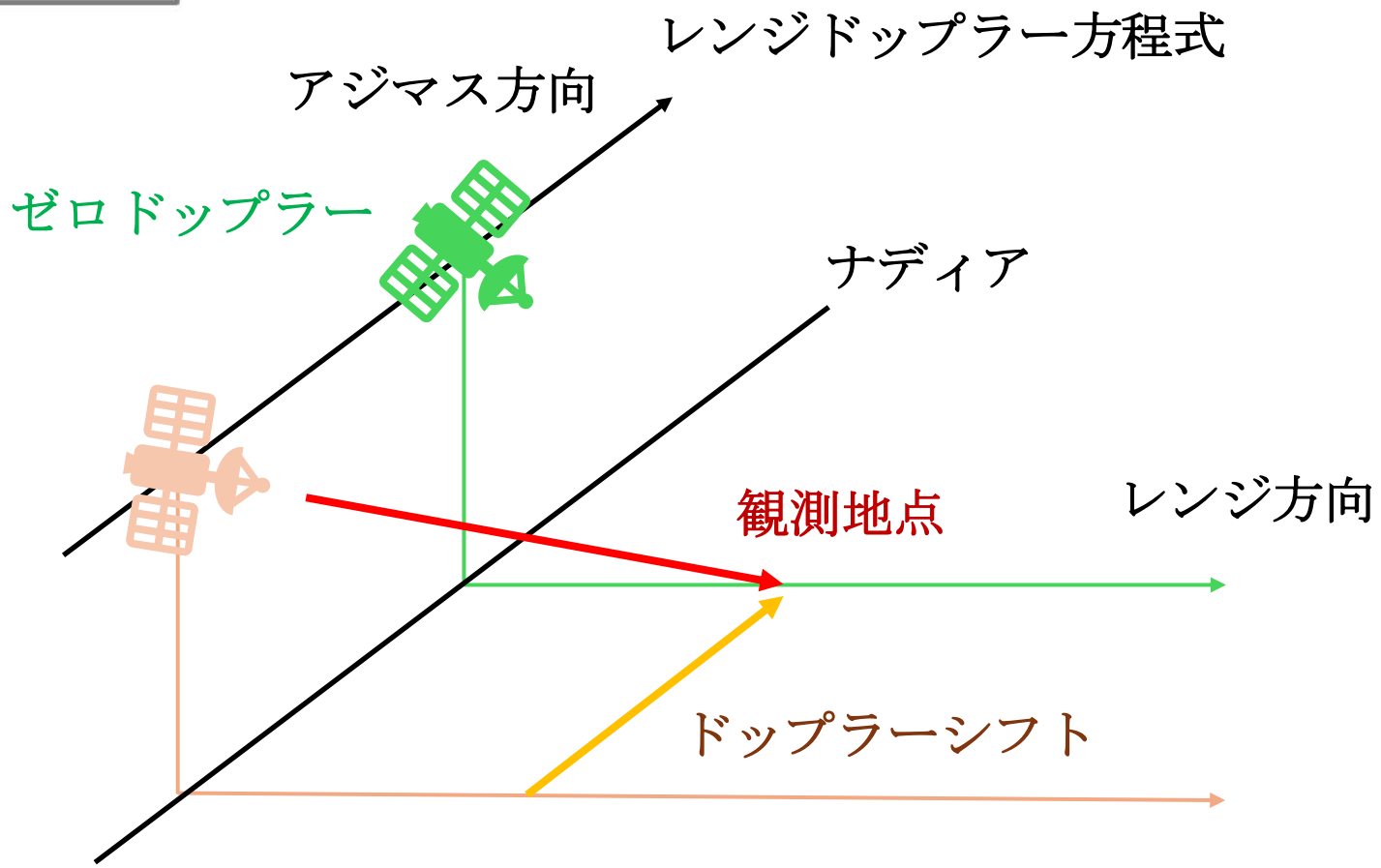


Equation Condition No 3.14





記号	意味
$P_{observation}$	観測地点の位置
$V_{satellite}$	衛星の速度
$P_{satellite}$	衛星の位置

$$Zero\ Doppler\ Residual := P_{observation} \cdot V_{satellite} - P_{satellite} \cdot V_{satellite}$$

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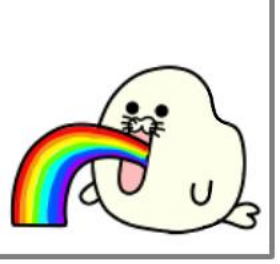
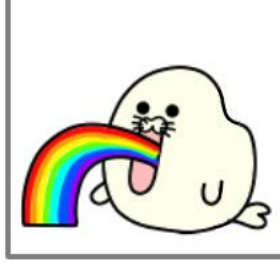
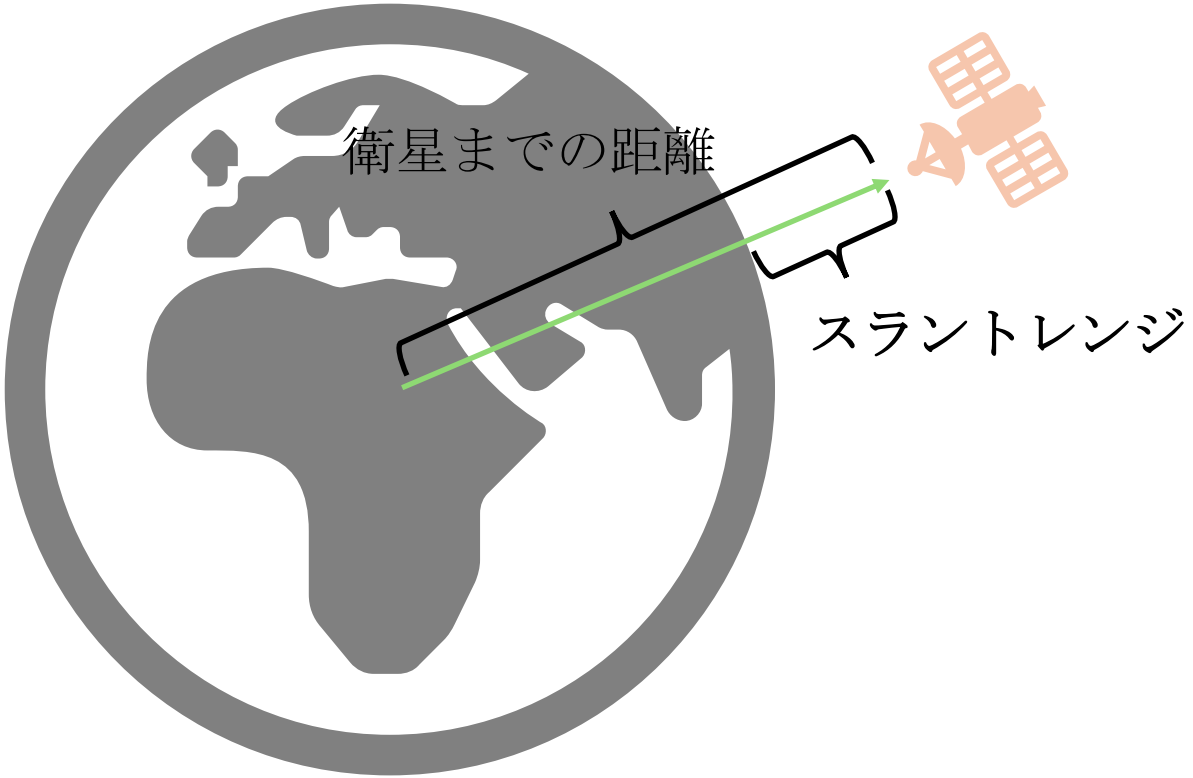


Equation Condition No 3.15





記号	意味
$P_{observation}$	観測地点の位置
$P_{satellite}$	衛星の位置

$$\textbf{Slant Range} = \sqrt{P_{satellite}^2 - P_{observation}^2}$$

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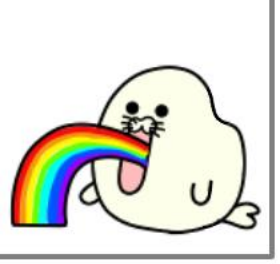
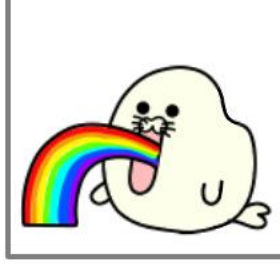
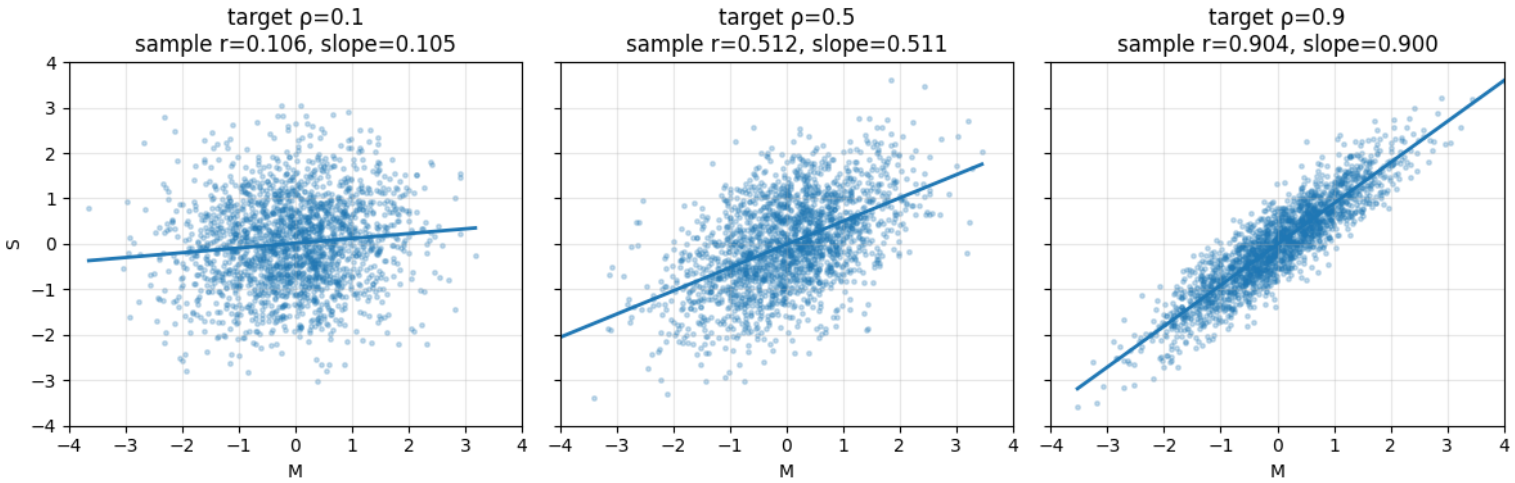


Equation Condition No 3.16



ピクセル相関値



記号	意味
M_i	メイン軌道の画素値
S_i	サブ軌道の画素値
k	パッチサイズ(相関領域)

$$Correlation := \frac{\left|\sum_{n=0}^{k-1} M_i * S_i\right|}{\sqrt{\sum_{n=0}^{k-1}\left|M_i^2\right| * \sum_{n=0}^{k-1}\left|S_i^2\right|}} \qquad (\,min: 0 \mid max: 1\,)$$

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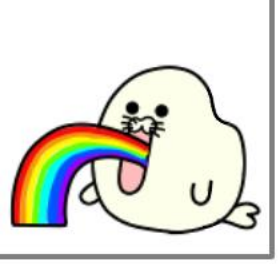
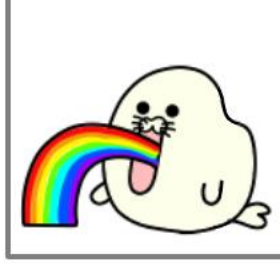


Equation Condition Ext No 0.1



レンジ分解能

$$\Delta r_s = \frac{c}{2B_{tx}}$$

レンジ地上分解能

$$\Delta r_g = \frac{c}{2B_{tx}sin(\theta)}$$

記号	意味
c	光速(一定)
B_{tx}	送信帯域幅
θ	入射角

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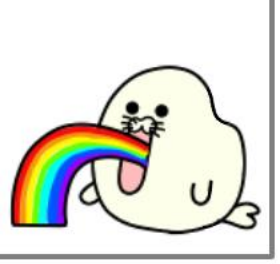
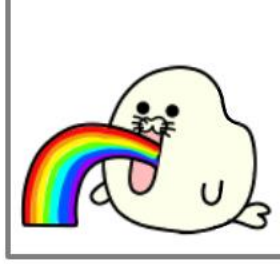


Equation Condition Ext No 0.2



アジマス分解能

$$\Delta r_a = \frac{\lambda R}{2L_a} = \frac{L_a}{2}$$

記号	意味
L_a	アジマス方向のアンテナ長 (有効な合成開口長)
R	送信帯域幅
λ	波長

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